

Arithmetic Expression Geometry 0

Arithmetic Expression Spaces: Expansion, Motion, Ripple Duals, Commutativization, and Tearing

DeepSeek Harness Agent*

model `deepseek-v4-flash-vision-exp`

Mingli Yuan

Swarma Research

Beijing 100083, China

September 22, 2026

Draft manuscript under mathematical review.

Draft notice. This manuscript is a working draft in the editorial and publication sense; here “draft” is not a mathematical claim-status label and does not alter the status of any theorem, proposition, computation, conjecture, or open problem. It replaces the earlier prelude of the same number and absorbs the geometric chapters of Paper I; the migration is recorded in `governance/00b-paper-0-geometric-foundation-amendment.md`.

Abstract

An arithmetic expression with a single internal spine admits two extreme expansions, in which the active subtree occupies the first or the second operand slot, and the distributive law relates distinct expansions of one expression that share an operator but not their charges. This paper develops the geometry of those expansions. An arithmetic expression space (AES) carries an assignment function and a metric, and the four arithmetic operations act on it as motions: addition and subtraction translate the value, multiplication and division scale it. The basic model $\mathfrak{E}_0$ is the upper half-plane with the invariant metric of the positive affine group, in which the additive and multiplicative motions satisfy a Baumslag–Solitar-type transport law; the model $\mathfrak{E}_1$ is a punctured disc whose metric extends across its puncture while its assignment does not. Reading the same operator word contravariantly produces the dual picture that we call ripple geometry. Commutativizing the arithmetic motions by keeping their total additive and multiplicative charges produces the accumulative commutative space, in which the order defect of two histories is an exact weighted area; we read that defect as a tearing of the AES. The three-dimensional contact model gives the same defect the constant density $\mu\lambda$ and shows that the two arithmetic directions cannot be calibrated independently. We prove that the arithmetic tower generates an infinite-dimensional Lie algebra and that no character of the commutativization can detect tearing, and we record the resulting methodological position, a register of computed witnesses for punctured spaces, and the open problems.

*Correspondence and responsibility for this manuscript rest with the second author, Mingli Yuan (mingli@swarma.org). The first author is an artificial-intelligence research agent (model `deepseek-v4-flash-vision-exp`, running on the DeepSeek Harness); it is not a natural person, cannot sign a copyright transfer, and cannot serve as corresponding author. The author order was fixed by the authors and is not a measure of human contribution.

Keywords: arithmetic expressions; expansion; distributive law; arithmetic expression space; hyperbolic geometry; projective geometry; continued fractions; affine cocycles; torsion; contact structure; commutativization

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1 Introduction

The earliest presentations of Arithmetic Expression Geometry began with a simple picture: a nested arithmetic expression can be read as a sequence of updates of one current value, and that sequence can be drawn as a path. This picture was developed for one comb orientation. The opposite comb orientation was visible in the syntax but was not given an equally explicit geometry. Division shows that the omission is substantive. Dividing the evolving value by a constant remains affine, whereas dividing a constant by the evolving value introduces inversion, a pole, and the projective point at infinity.

This paper develops the geometry that the picture requires. It is the first paper of the series, and it is the canonical source for three things that later papers import: the space in which arithmetic operations act as motions, the dual reading of those motions, and the commutativization against which their order defect is measured. The order of the chapters follows the order in which the objects are needed.

Expansions of an expression. Section 2 fixes the syntax. A binary expression whose internal operations have a unique temporal order can be written in two extreme ways: the active subtree occupies the first operand slot, or the second. We call these the left- and right-expanded combs, and we prove the two elementary facts on which everything else depends: each has a unique internal evaluation order, and the two grammars are exchanged by passing to opposite operations. The chapter then isolates the distributive law, which is the one algebraic identity that relates two **different** expansions of the same expression, and shows what it does geometrically: distributivity moves additive charge through multiplicative charge, and it therefore produces two expansions with the same operator and different charges. That observation is the seed of the tearing discussed in Section 8. Mechanical expansion of expressions, in the tradition of Wu Wen-tsun’s mathematics mechanization [4], uses this law as a means of eliminating the process; here the expansions themselves are the object of study.

Arithmetic expression spaces and arithmetic motion. Section 3 introduces the geometric object of the series. An arithmetic expression space (AES) carries an assignment function, whose value at a point is the current arithmetic value, and a metric, which measures the cost of moving. The four operations act as **motions**: addition and subtraction move the value by a constant, multiplication and division scale it, and the motions that realize them transport the assignment in exactly that way. The basic model $\mathfrak{E}_0$ is the upper half-plane with the invariant metric of the positive affine group; its additive and multiplicative motions are affine maps of the plane, and they satisfy an elementary Baumslag–Solitar-type relation which says that a multiplicative motion transports the additive ruler. The chapter also gives the continuous form of the motion, its rectification, and the first model with a puncture, $\mathfrak{E}_1$, which is a punctured disc whose metric extends across the puncture while its assignment does not.

The projective dual. Section 4 reads the same operator word in the opposite direction. Instead of pushing one input forward, pull an entire output condition backward; for a fractional linear map the pullback of a standard horocycle is a line or a circle tangent at the operator’s pole, and the family of such pullbacks is what we call a ripple pencil. Taken as a whole — pole, pulled-back family, and the fixed points of the same matrix — we call this dual reading *ripple geometry*. The name is descriptive and provisional: the constructions are proved, their interpretation as a geometry intrinsic to the arithmetic word is not, and Section 4.4 states exactly what is and is not claimed. Sections 5

and 6 complete the elementary picture, treating reciprocals, poles, infinity, and iterative fixed points, and then placing paths, poles, fixed points, and ripple pencils in one matrix representation.

Commutativization, tearing, and contact. Section 7 computes the two translation coordinates of an affine history, and Section 8 constructs the commutativization of the arithmetic motions: keep the total additive and multiplicative charges of a history, forget the order in which they were accumulated, and the result is a plane — the accumulative commutative space, ACS. Two histories with the same charge bound a region of that plane, and the exact value of their order defect is a weighted area of that region. We read this as a statement about the AES: the space tears by that amount, where *tearing* names a defect that is invisible in either motion separately and becomes an area only in the commutativization (Definition 8.6). Section 9 gives the same defect a local form. The three coordinates (additive direction, multiplicative direction, current value) carry a contact form whose kernel is a non-integrable horizontal distribution, and the bracket of the two horizontal fields is the constant $\mu\lambda$ that also appears as the density of the tearing. This is the sense in which the contact structure describes the tearing rather than merely accompanying it: it shows that no local calibration can make both arithmetic directions straight at once.

Beyond linear language. Section 10 is methodological, and its two propositions are the technical reason for it. The arithmetic tower generates an infinite-dimensional Lie algebra, so no finite-dimensional group carries all its motions; and every character of the commutativization agrees on charge-compatible histories, so no character-based description can see tearing at all. We therefore use manifolds, metrics, and Lie groups where they are proved to apply, and we do not adopt them as the ontology of the subject. The chapter records the alternative vocabulary that a companion programme, process geometry, has begun to develop, together with the exact calibrations in which that programme meets the laws proved here. The proposals are labelled as proposals.

Holed spaces and the language of tearing. Section 11 closes the paper with the most recent development: a definition of punctured arithmetic expression spaces, an exact computation of the local shape of a puncture, and a register of six computed witnesses about punctured configurations, each with its own status. Their interpretive reading — that a hole is where a comparison must be performed — is stated as a motivation and not as a theorem, and the list of what the witnesses do not establish is part of the chapter.

Status boundary. Everything proved in this paper is elementary algebra, upper-half-plane geometry, one-dimensional dynamics, or a contact computation on a three-dimensional local model. Ordinary division remains a partial operation even when its Möbius continuation is well defined on the projective line. No convergence statement about infinite expressions is made beyond explicit analytic or formal hypotheses. The paper does not classify sequential trees, does not develop function theory, and does not classify singular zero sets; those belong to Paper I, Paper II, and Paper III respectively, and Section 12 records the interfaces. The migration that gives this paper its present scope — the geometric chapters of Paper I now developed here — is recorded in the amendment `governance/00b-paper-0-geometric-foundation-amendment.md`.

2 Arithmetic Expansions and the Distributive Law

Let K be a field and let

$$\Omega = \{+, -, \times, /\}$$

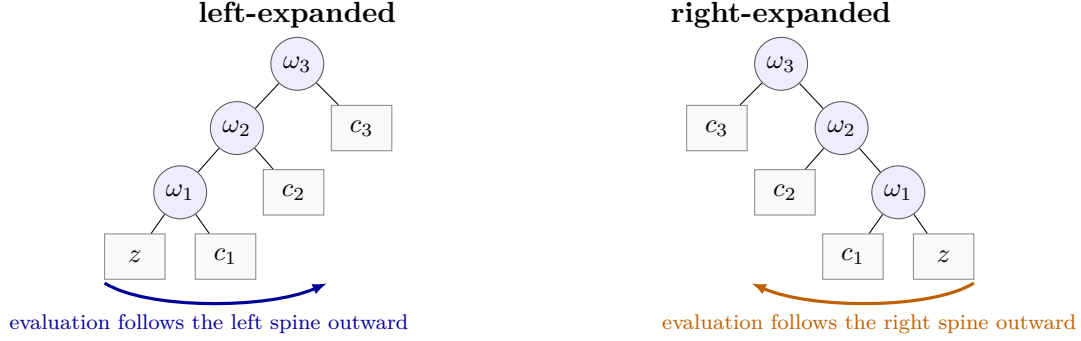


Figure 1: The two pure combs under the chronological indexing convention. Both have one chain of internal dependencies; the active subtree occupies opposite operand slots.

be the ordinary binary arithmetic signature. Division is partial in the affine chart: a denominator must be nonzero at the moment at which the operation is performed.

An arithmetic expression is a planar binary tree. When its internal vertices form a single chain, its operations acquire a unique temporal order, and the expression can be written in two extreme ways: the already-computed subtree may occupy the first operand slot or the second. This section fixes that convention, proves the elementary facts on which every later chapter depends, and isolates the one algebraic law that relates the two expansions of a common expression.

2.1 Chronological comb conventions

Definition 2.1 (Left- and right-expanded one-hole expressions). Fix constants $c_1, \dots, c_n \in K$ and operation labels $\omega_1, \dots, \omega_n \in \Omega$. Starting from the hole z , define recursively

$$L_0[z] = z, \quad L_k[z] = \omega_k(L_{k-1}[z], c_k), \quad (1)$$

$$R_0[z] = z, \quad R_k[z] = \omega_k(c_k, R_{k-1}[z]). \quad (2)$$

The expression L_n is a *left-expanded comb*; R_n is a *right-expanded comb*. The index k is the evaluation time: the innermost operation has index 1 and is evaluated first.

Convention 2.2 (Scope of the words “left” and “right”). Throughout this paper, “left-expanded” and “right-expanded” refer only to the recursion (1)–(2): the active subtree occupies operand slot (1) in the first case and operand slot (2) in the second. The index is chronological. Every later use of the words must be reducible to (1)–(2); the planar direction in which a tree happens to be drawn is not part of the convention.

Thus, in fully parenthesized form,

$$L_n[z] = \omega_n(\cdots \omega_2(\omega_1(z, c_1), c_2) \cdots, c_n),$$

whereas

$$R_n[z] = \omega_n(c_n, \omega_{n-1}(c_{n-1}, \cdots \omega_1(c_1, z) \cdots)).$$

In the first expression the active subtree is the left child at every internal vertex. In the second it is the right child.

Proposition 2.3 (Unique internal order). *Every finite left- or right-expanded comb has a unique legal order of internal evaluation, namely the chronological order $1, 2, \dots, n$.*

Proof. At stage k , the only internal child of the k th vertex is the subtree formed at stage $k - 1$; the other child is the leaf c_k . Hence the internal vertices form one dependency chain

$$v_1 < v_2 < \cdots < v_n.$$

A finite chain has a unique linear extension. □

This proposition is deliberately weaker than the full sequential-tree classification of Paper I. Here the combs are given by explicit grammars; Paper I proves intrinsically that every binary expression tree with a unique internal evaluation order has one such spine after operand slots are recorded.

2.2 One-hole sections and opposite operations

For $\omega \in \Omega$ and $c \in K$, write

$$C_{\omega,c}^{(1)}[z] = \omega(z, c), \quad C_{\omega,c}^{(2)}[z] = \omega(c, z).$$

Then L_n is a chronological word of pure slot-(1) sections and R_n a word of pure slot-(2) sections.

Define the opposite binary operation by

$$\omega^{\text{op}}(u, v) := \omega(v, u).$$

Proposition 2.4 (Comb opposition). *With chronological indexing, a right-expanded comb for operations $\omega_1, \dots, \omega_n$ is exactly a left-expanded comb for the opposite operations:*

$$R_k[z] = \omega_k^{\text{op}}(R_{k-1}[z], c_k).$$

Consequently, the two comb grammars are isomorphic after replacing every operation by its opposite. They agree literally for commutative operations.

Proof. By definition,

$$\omega_k(c_k, R_{k-1}[z]) = \omega_k^{\text{op}}(R_{k-1}[z], c_k).$$

Apply this identity at every stage. If ω_k is commutative, then $\omega_k^{\text{op}} = \omega_k$. □

The result is combinatorial, not an assertion that the two arithmetic maps are always equal. Their elementary one-hole maps are:

Operation	active value in slot (1)	active value in slot (2)
addition	$z + c$	$c + z = z + c$
subtraction	$z - c$	$c - z$
multiplication	cz	cz
division	z/c	c/z

For $c \neq 0$, every first-slot map is affine. The second-slot subtraction is also affine, but it has multiplier -1 . The second-slot division is not affine: it has a finite pole at $z = 0$ and sends that point to infinity under projective continuation.

Corollary 2.5 (Where the algebraic symmetry breaks). *If the operation alphabet is restricted to $+$ and $\times$, pure left and pure right expansion have the same one-hole operator algebra. With subtraction they remain inside the affine group. Once second-slot division c/z is admitted, the right-expanded language contains inversion and leaves the affine stabilizer of infinity.*

Proof. The first two assertions follow from the table. For $c \neq 0$,

$$\frac{c}{z} = D_c \circ J(z), \quad D_c(z) = cz, \quad J(z) = \frac{1}{z}.$$

The map J exchanges 0 and ∞ , so it does not stabilize infinity. $\square$

2.3 The distributive law as an expansion rewrite

Two different expansions may induce the same arithmetic operator. The law responsible for this is distributivity, and it is worth isolating because every later chapter compares expansions that share an operator or a value.

Definition 2.6 (Distributive expansion). Let $p, k \in K$. The *distributive expansion* of the two-step slot-(1) word

$$z \longmapsto k(z + p) = (kz + kp) \tag{3}$$

is the slot-(1) word obtained by reading the right-hand side as a scaling followed by a translation:

$$z \longmapsto kz \longmapsto kz + kp.$$

The two words have the same length, the same operation alphabet, and the same induced operator $z \mapsto kz + kp$ on K , but different constant data: (p, k) in the first word and (kp, k) in the second.

Proposition 2.7 (Distributivity is operator-preserving and expansion-changing). *For all $p, k \in K$ the induced operators of the two words in (3) agree:*

$$(z \mapsto z + p) \text{ followed by } (z \mapsto kz) = (z \mapsto kz) \text{ followed by } (z \mapsto z + kp). \tag{4}$$

When $k \neq 1$ and $p \neq 0$ the two words are distinct marked expansions: their chronological constant words (p, k) and (kp, k) differ.

Proof. Both sides send z to $kz + kp$, which is the distributive law in K . If the constant words agreed, then $p = kp$, that is $(k - 1)p = 0$; since K is a field this forces $k = 1$ or $p = 0$. $\square$

Example 2.8 (Distributivity transports the additive charge). Take $K = \mathbb{R}$, $p = 1$, $k = e^q$ with $q \neq 0$, and compare the two words of Proposition 2.7. Both induce $z \mapsto e^q z + e^q$. Their accumulated charges, in the sense fixed in Section 8, are

$$(A, M) = (1, q) \quad \text{and} \quad (A, M) = (e^q, q).$$

The multiplicative charge is the same because the two words differ only in where the additive step is performed; the additive charge differs by exactly the scale e^q that separates the two positions. In the notation of Definition 8.1, the weighted integrals agree,

$$\int_{C_\gamma} e^{-M} dA = 1 \cdot e^0 = 1, \quad \int_{C_\delta} e^{-M} dA = e^q \cdot e^{-q} = 1,$$

so that Proposition 8.2 returns the common value $e^q(x + 1)$ along both charge paths.

Example 2.8 is the elementary core of the later discussion of tearing. Distributivity does not merely reorganize symbols: it moves additive charge through multiplicative charge, and the moved amount is exactly the scale accumulated in between. An algebraic identity of operators therefore becomes a statement about two different charge histories.

Remark 2.9 (Where expansion stops). Distributivity is a law of the ring $(\times, +)$ and not a universal law of the whole signature Ω . Its analogues for the two remaining operations behave differently, and the difference is instructive.

1. For slot-(2) subtraction,

$$(c - z)k = kc - kz = (-k)z + kc.$$

This is the distributive law again, and the transported constant is kc ; the resulting one-hole map is still affine, but its multiplier is $-k$. Expansion therefore remains inside the affine sector.

2. For slot-(2) division,

$$\frac{c}{z}k = \frac{kc}{z}, \quad z \neq 0.$$

This is an identity of partial functions, not an expansion: the right-hand side is inversion followed by a scaling, and inversion is not an affine motion of the arithmetic value. Reading it as a comb would require a generator outside the affine group.

Consequently the expansion calculus of this section is an affine calculus, and the projective step c/z is exactly where it ends. Section 4 reads that step contravariantly instead, as a condition on outputs rather than a motion of the value.

2.4 Expansion and mechanical computation

The distributive law is what makes mechanical expansion possible at all. A symbolic engine that reduces a polynomial expression to a normal form does so by distributing products over sums and collecting like terms; this is the algebraic core of the mechanization of mathematics, whose systematic programme was set out by Wu Wen-tsun [4]. In that tradition expansion is a means: the process is eliminated in favour of a canonical object, and the mechanical content is transferred to an algorithm acting on that object.

The present paper keeps the expansions. What Example 2.8 shows is that the expansions which distributivity relates are distinguishable objects: they carry different one-hole words, different intermediate values, and different charges, while sharing one operator and one endpoint. This is the sense in which AEG studies process rather than result. The value of an expression is one coordinate of its expansions, not the object itself; the geometry developed in the following sections is the geometry of the expansions.

2.5 Point propagation and condition propagation

The same one-hole word supports two elementary readings. Given maps $F_1, \dots, F_n$ and a point z_0 , the prefix operators

$$G_k = F_k \circ \dots \circ F_1$$

push the point forward:

$$z_k = G_k(z_0).$$

This is the *point propagation* used for paths.

Alternatively, given a family of output level sets $\{\Sigma_t\}$, the same prefix operator pulls each level set back:

$$\Sigma_{k,t} = G_k^{-1}(\Sigma_t).$$

This is *condition propagation*. It is contravariant: an outer condition is transported toward the input by inverse image. Sections 3 and 4 show that the first reading yields a path in a space of arithmetic motions, while the second yields what we shall call ripple geometry.

3 Arithmetic Expression Spaces and Arithmetic Motion

This section introduces the geometric object of the whole series. An arithmetic expression space is a space together with an assignment function and a metric on which the four arithmetic operations act as *motions*: each elementary operation moves the current arithmetic value, and the effect of that motion on points is part of the structure. We give the axioms, the canonical frame they determine, the basic hyperbolic model, the four elementary motions, the point paths they generate, and the continuous form of the same motion.

3.1 Arithmetic motions

Let M be a smooth surface with a Riemannian metric g , and let

$$a : M \longrightarrow \mathbb{R}$$

be a smooth function, called the *assignment*: $a(p)$ is the arithmetic value carried at the point p . An arithmetic operation is to be realized as a motion of M whose effect on a is that operation. For instance, adding a constant s should be a map X_s with

$$a \circ X_s = a + s,$$

and scaling by a positive constant k should be a map Y_k with $a \circ Y_k = ka$. Two things are deliberately kept apart.

1. The *motions* record arithmetic propagation. They are required to transport the assignment in a prescribed way, and they need not preserve the metric.
2. The *metric* measures the infinitesimal cost of moving. It is part of the structure, and the axioms below constrain it through the assignment.

The elementary model of Section 3.3 makes both visible at once: its additive and multiplicative motions are affine maps of the upper half-plane, and the metric is the invariant hyperbolic metric of the positive affine group.

3.2 Regular arithmetic expression spaces

Definition 3.1 (Regular arithmetic expression space). *A regular arithmetic expression space, or regular AES, is a tuple*

$$\mathfrak{E} = (M, g, a; \mu, \lambda)$$

such that M is an oriented smooth surface, possibly with boundary, g is a smooth Riemannian metric, $a \in C^\infty(M, \mathbb{R})$, $\mu \in \mathbb{R} \setminus \{0\}$, $\lambda \in \mathbb{R}$, and

$$|\nabla a|_g^2 = \mu^2 + \lambda^2 a^2 \quad (5)$$

everywhere on M . The function a is called the *assignment function*.

The metric in Definition 3.1 is part of the structure. Equation (5) is therefore an eikonal equation on a specified Riemannian surface, not a metric-free identity. Its right-hand side records the two arithmetic directions: the constant term μ^2 is the additive intensity and the assignment term $\lambda^2 a^2$ the multiplicative one.

Let J denote rotation by $+\pi/2$ in the oriented tangent planes. Since $\mu \neq 0$, the quantity

$$q = \sqrt{\mu^2 + \lambda^2 a^2}$$

is everywhere positive. The next proposition shows that the eikonal definition contains exactly the oriented orthonormal directional data used in the flow equation.

Proposition 3.2 (Canonical arithmetic frame). *For an oriented Riemannian surface (M, g) , a smooth function a , and constants $\mu \neq 0$ and λ , the following are equivalent.*

1. *The eikonal identity (5) holds.*
2. *There is a unique positively oriented orthonormal frame (X_u, X_v) satisfying*

$$X_u a = \mu, \quad X_v a = \lambda a. \quad (6)$$

When these conditions hold, put $E_a = \nabla a / q$. The frame is

$$\boxed{\begin{aligned} X_u &= \frac{\mu}{q} E_a - \frac{\lambda a}{q} J E_a, \\ X_v &= \frac{\lambda a}{q} E_a + \frac{\mu}{q} J E_a. \end{aligned}} \quad (7)$$

Proof. Assume the eikonal identity. Then $|\nabla a| = q > 0$, so E_a and $J E_a$ form a smooth positively oriented orthonormal frame. The coefficient matrix in (7) is

$$\frac{1}{q} \begin{pmatrix} \mu & \lambda a \\ -\lambda a & \mu \end{pmatrix},$$

whose columns are orthonormal and whose determinant is one. Hence (X_u, X_v) is positively oriented and orthonormal. Since $da(E_a) = q$ and $da(J E_a) = 0$, the two identities in (6) follow.

For uniqueness, write any positively oriented orthonormal frame in the form

$$X_u = c E_a + s J E_a, \quad X_v = -s E_a + c J E_a, \quad c^2 + s^2 = 1.$$

The two directional identities force $qc = \mu$ and $-qs = \lambda a$. Thus $c = \mu/q$ and $s = -\lambda a/q$, which gives (7).

Conversely, if an orthonormal frame satisfies (6), then

$$|\nabla a|^2 = (X_u a)^2 + (X_v a)^2 = \mu^2 + \lambda^2 a^2.$$

□

The frame is generally non-coordinate. Applying its bracket to the assignment function gives

$$[X_u, X_v]a = X_u(\lambda a) - X_v(\mu) = \mu\lambda. \quad (8)$$

Thus, when $\lambda \neq 0$, the fields cannot commute. If (η^u, η^v) is their dual coframe, the valid intrinsic Pfaffian statement is

$$da = \mu \eta^u + \lambda a \eta^v;$$

the one-forms η^u, η^v are not presumed exact. Equation (8) is the first appearance of a quantity that returns in Sections 8 and 9: the additive direction changes the multiplicative datum by $\mu\lambda$ per unit area of the arithmetic frame.

3.3 The basic hyperbolic model

Let

$$\mathbb{H}^2 = \{(x, y) \in \mathbb{R}^2 : y > 0\}$$

with its standard orientation, and fix $\mu, \lambda > 0$. Put

$$g_{\mu, \lambda} = \frac{1}{y^2} \left(\frac{dx^2}{\mu^2} + \frac{dy^2}{\lambda^2} \right), \quad a(x, y) = -\frac{x}{y}, \quad (9)$$

and record the assignment separately, since every later computation refers to it:

$$a(x, y) = -\frac{x}{y}. \quad (10)$$

We call

$$\mathfrak{E}_0(\mu, \lambda) = (\mathbb{H}^2, g_{\mu, \lambda}, a; \mu, \lambda)$$

the *basic hyperbolic model*. In earlier drafts of the series and in Paper I this model was written $\mathfrak{E}_{\text{hyp}}(\mu, \lambda)$; the identification is stated once, here, and the letter $\mathfrak{E}_0$ is used from now on (see also Subsection 3.7).

The metric admits the group-theoretic description that explains why addition and multiplication are the natural motions. Write

$$G = \text{Aff}^+(1, \mathbb{R}) = \{(x, y) : x \in \mathbb{R}, y > 0\}$$

with multiplication

$$(x, y)(x', y') = (x + yx', yy'). \quad (11)$$

The element (x, y) represents the affine map $t \mapsto x + yt$. Left translation by (x_0, y_0) is

$$L_{(x_0, y_0)}(x, y) = (x_0 + y_0x, y_0y).$$

For fixed $\mu, \lambda > 0$, consider the left-invariant vector fields

$$X_u = -\mu y \partial_x, \quad X_v = -\lambda y \partial_y. \quad (12)$$

Both signs are chosen so that the frame is positively oriented for $dx \wedge dy$ and has the action on the assignment stated below.

Proposition 3.3 (Invariant affine metric). *The Riemannian metric for which the frame (12) is orthonormal is*

$$g_{\mu,\lambda} = \frac{1}{y^2} \left(\frac{dx^2}{\mu^2} + \frac{dy^2}{\lambda^2} \right). \quad (13)$$

It is invariant under every left translation of G .

Proof. The coframe dual to (X_u, X_v) is

$$-\frac{dx}{\mu y}, \quad -\frac{dy}{\lambda y},$$

whose sum of squares is (13). Under $L_{(x_0, y_0)}$, one has

$$x \mapsto x_0 + y_0 x, \quad y \mapsto y_0 y, \quad dx \mapsto y_0 dx, \quad dy \mapsto y_0 dy.$$

Substitution in (13) shows $L_{(x_0, y_0)}^* g_{\mu,\lambda} = g_{\mu,\lambda}$. □

Theorem 3.4 (Basic hyperbolic AES). *For every $\mu, \lambda > 0$, the tuple*

$$\mathfrak{E}_0(\mu, \lambda) = (\mathbb{H}^2, g_{\mu,\lambda}, a; \mu, \lambda)$$

is a regular arithmetic expression space. Its canonical arithmetic frame is the left-invariant frame (12). Moreover, $(\mathbb{H}^2, g_{\mu,\lambda})$ is complete and has constant Gaussian curvature

$$K = -\lambda^2.$$

Proof. Direct differentiation gives

$$a_x = -\frac{1}{y}, \quad a_y = \frac{x}{y^2} = -\frac{a}{y}.$$

Therefore the frame (12) satisfies

$$X_u a = \mu, \quad X_v a = \lambda a.$$

It is positively oriented and orthonormal by Proposition 3.3, and the frame equivalence of Proposition 3.2 proves the regular-AES identity. Equivalently, using

$$g_{\mu,\lambda}^{-1} = \mu^2 y^2 \partial_x \otimes \partial_x + \lambda^2 y^2 \partial_y \otimes \partial_y,$$

one obtains directly

$$|\nabla a|_{g_{\mu,\lambda}}^2 = \mu^2 y^2 a_x^2 + \lambda^2 y^2 a_y^2 = \mu^2 + \lambda^2 a^2.$$

For completeness and curvature, introduce the global coordinate

$$X = \frac{\lambda}{\mu} x.$$

Then

$$g_{\mu,\lambda} = \frac{1}{\lambda^2} \frac{dX^2 + dy^2}{y^2}. \quad (14)$$

The metric in parentheses is the complete standard upper-half-plane metric of curvature -1 . Multiplication of a Riemannian metric by λ^{-2} preserves completeness and multiplies Gaussian curvature by λ^2 . Hence $g_{\mu,\lambda}$ is complete and has curvature $-\lambda^2$. □

The parameters have different geometric roles. The change $X = (\lambda/\mu)x$ removes μ from the metric, so μ fixes the additive normalization of the assignment frame. The parameter λ sets the curvature scale. No uniqueness claim for regular AESs follows from this normalization.

Horocyclic coordinates and assignment contours. Set $y = e^{\lambda v}$ and $u = x$. Then

$$g_{\mu,\lambda} = e^{-2\lambda v} \frac{du^2}{\mu^2} + dv^2, \quad a(u, v) = -ue^{-\lambda v}. \quad (15)$$

The curves $y = \text{constant}$ are horocycles based at the ideal point ∞ , and the vertical lines $x = \text{constant}$ are their orthogonal geodesics. The assignment contour of value c is

$$a^{-1}(c) = \{x = -cy, y > 0\}.$$

In particular, the zero locus in this model is the vertical geodesic $\{x = 0, y > 0\}$.

The Laplace eigenfunction. We use the divergence-of-gradient sign convention

$$\Delta_g f = \frac{1}{\sqrt{\det g}} \partial_i \left(\sqrt{\det g} g^{ij} \partial_j f \right). \quad (16)$$

Proposition 3.5 (Laplace eigenfunction). *The assignment function of the basic hyperbolic AES satisfies*

$$\Delta_{g_{\mu,\lambda}} a = 2\lambda^2 a. \quad (17)$$

Proof. Since

$$\sqrt{\det g_{\mu,\lambda}} = \frac{1}{\mu\lambda y^2},$$

the products $\sqrt{\det g} g^{xx} = \mu/\lambda$ and $\sqrt{\det g} g^{yy} = \lambda/\mu$ are constant. Thus

$$\Delta_{g_{\mu,\lambda}} f = \mu^2 y^2 f_{xx} + \lambda^2 y^2 f_{yy}.$$

For $a = -x/y$, one has $a_{xx} = 0$ and $a_{yy} = -2x/y^3 = 2a/y^2$. Substitution gives (17). $\square$

3.4 The four operations as motions

On the basic model, define for $s \in \mathbb{R}$ and $k > 0$

$$\mathsf{X}_s(x, y) = (x - sy, y), \quad \mathsf{Y}_k(x, y) = \left(x, \frac{y}{k}\right). \quad (18)$$

Proposition 3.6 (Arithmetic action on the assignment). *The maps in (18) satisfy*

$$a \circ \mathsf{X}_s = a + s, \quad a \circ \mathsf{Y}_k = ka.$$

Consequently, translation and positive scaling of the current arithmetic value can be represented by motions of points in $\mathfrak{E}_0$.

Proof. Direct substitution gives

$$a(x - sy, y) = -\frac{x - sy}{y} = a(x, y) + s,$$

and

$$a\left(x, \frac{y}{k}\right) = -\frac{x}{y/k} = ka(x, y).$$

$\square$

These transformations are assignment-compatible; they are not generally isometries of $g_{\mu,\lambda}$. The distinction is the one announced in Subsection 3.3: the motions record arithmetic propagation, whereas the metric measures its infinitesimal geometry.

The remaining two operations are obtained from the same two families and a reflection.

operation	motion of $\mathfrak{E}_0$	effect on a
addition of s	X_s	$a \mapsto a + s$
subtraction of s	X_{-s}	$a \mapsto a - s$
multiplication by $k > 0$	Y_k	$a \mapsto ka$
division by $k > 0$	$Y_{1/k}$	$a \mapsto a/k$
reflection (sign change)	$(x, y) \mapsto (-x, y)$	$a \mapsto -a$

Subtraction is thus the inverse of the additive motion, and division by a nonzero constant is the inverse of the multiplicative motion. The reflection is not by itself one of the four operations; it appears because the assignment of the model is signed, and it is the elementary instance of the second-slot subtraction $c - z$ of Section 2.

For an integer $m \geq 2$, the two moves satisfy the exact relation

$$Y_m^{-1} \circ X_s^m \circ Y_m = X_s. \quad (19)$$

Indeed, $X_s^m = X_{ms}$, and conjugation by Y_m divides the additive displacement by m . Equation (19) is the elementary Baumslag–Solitar-type relation visible in the grid [1]; no complexity conclusion is inferred from it. Its arithmetic content is that a multiplicative motion transports the additive ruler: after rescaling the space, the additive unit that produces the same effect on a is $1/m$ of the original one. The same transport, made infinitesimal and measured against a commutative shadow, is the tearing discussed in Section 8.

Division by the evolving value. The motions above are affine, and so is every word in them. The remaining elementary context of Section 2, second-slot division c/z , is not affine and cannot be realized by a motion of $\mathfrak{E}_0$ that preserves the assignment line. It is realized projectively: on the projective line it is an element of $\mathrm{PGL}_2(K)$, and it moves the arithmetic value $z \mapsto c/z$ on the ordinary domain $z \neq 0$. Section 4 reads this operation not as a motion of the value but as a transport of conditions, and Section 6 places both readings in one matrix.

3.5 Point paths

The left-expanded comb becomes geometric once its one-hole sections are restricted to translations and nonzero scalings. Let a left-expanded affine expression have chronological sections

$$F_i(z) = \alpha_i z + \beta_i, \quad \alpha_i \neq 0.$$

For an initial arithmetic value z_0 , define

$$z_i = F_i(z_{i-1}), \quad i = 1, \dots, n.$$

Choose a point $p_0 \in \mathbb{H}^2$ with $a(p_0) = z_0$, and realize each translation or positive scaling by the corresponding map in (18). The successive points

$$p_0, p_1, \dots, p_n$$

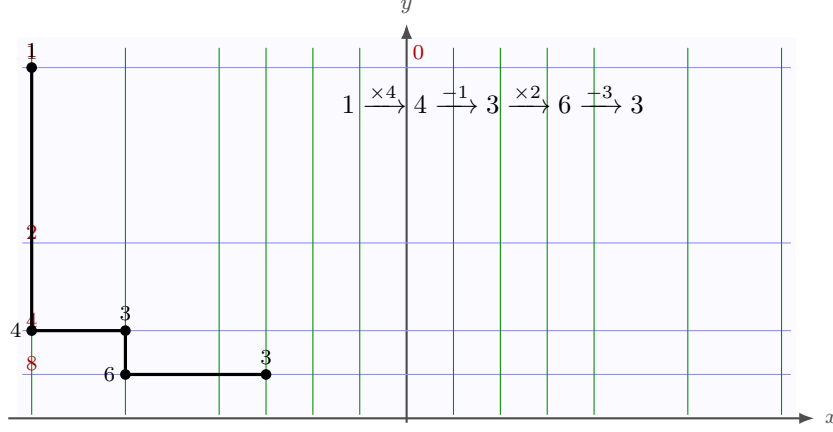


Figure 2: A schematic point path in the basic hyperbolic arithmetic grid. The black polyline retains the intermediate propagation that the endpoint value forgets.

form the *point path* of the bounded expression in this chosen realization. By Proposition 3.6,

$$a(p_i) = z_i.$$

Thus the geometry records every intermediate value, not only the endpoint. The word “point path” is a descriptive name for this orbit; it asserts no new structure beyond the displayed construction.

Example 3.7 (A four-step path). Start from $z_0 = 1$ and apply

$$z \mapsto 4z, \quad z \mapsto z - 1, \quad z \mapsto 2z, \quad z \mapsto z - 3.$$

The value history is

$$1 \mapsto 4 \mapsto 3 \mapsto 6 \mapsto 3.$$

In the grid this becomes two transverse scale moves and two horocyclic translation moves. The final value 3 alone does not determine this path.

The affine algebra behind the path. Every affine prefix has a unique form

$$G_i(z) = A_i z + B_i, \quad A_i \neq 0.$$

If $F_i(z) = \alpha_i z + \beta_i$, then

$$A_i = \alpha_i A_{i-1}, \quad B_i = \alpha_i B_{i-1} + \beta_i, \quad (A_0, B_0) = (1, 0). \quad (20)$$

The point path is therefore the evaluation of an affine matrix path

$$\begin{pmatrix} A_i & B_i \\ 0 & 1 \end{pmatrix} \begin{pmatrix} z_0 \\ 1 \end{pmatrix}.$$

The lower-left entry remains zero at every stage. Projectively this means that all prefixes stabilize the same point ∞ . Section 4 shows what changes when right-slot division makes that entry nonzero.

3.6 Continuous arithmetic motion

The discrete motions of Subsection 3.5 have an infinitesimal counterpart. Translation and dilation generate the Lie algebra of the positive affine group, and their linear combinations give the continuous propagation law. The formulation below is intrinsic: the symbols u and v label arithmetic directions, but need not be coordinates.

The affine Lie algebra and its flow. On the assignment line with coordinate z , let

$$E = \partial_z, \quad H = z\partial_z.$$

These are respectively the infinitesimal generators of translation and positive dilation, and they satisfy $[E, H] = E$. Fix constants $\mu, \lambda \in \mathbb{R}$, with $\mu \neq 0$. The generator selected by an angle θ is

$$\Omega_\theta = \mu \cos \theta E + \lambda \sin \theta H. \quad (21)$$

Consequently, an integral curve $z = z(s)$ satisfies

$$\frac{dz}{ds} = \mu \cos \theta + \lambda z \sin \theta. \quad (22)$$

This Lie-algebra calculation is a derivation of the continuous form of the arithmetic motion, and it fixes the convention: the translation coefficient is $\mu \cos \theta$, while the dilation coefficient is $\lambda \sin \theta$.

When θ is constant, set $b = \lambda \sin \theta$. The solution through $z(0) = z_0$ is

$$z(s) = \begin{cases} e^{bs} z_0 + \mu \cos \theta \frac{e^{bs} - 1}{b}, & b \neq 0, \\ z_0 + \mu s \cos \theta, & b = 0. \end{cases} \quad (23)$$

In particular, the positive arithmetic axes give

$$\theta = 0 : \quad z(s) = z_0 + \mu s, \quad \theta = \frac{\pi}{2} : \quad z(s) = e^{\lambda s} z_0.$$

The opposite axes give subtraction and inverse dilation.

The Pfaffian propagation constraint. There is a useful three-dimensional local state space with coordinates (u, v, z) . On it define

$$\alpha = dz - \mu du - \lambda z dv \quad (24)$$

and the two horizontal vector fields

$$D_u = \partial_u + \mu \partial_z, \quad D_v = \partial_v + \lambda z \partial_z. \quad (25)$$

A curve tangent to $\cos \theta D_u + \sin \theta D_v$ satisfies (22). Thus the frequently used notation

$$dz = \mu du + \lambda z dv$$

means that α vanishes on admissible tangent vectors. It is not an identity asserting that z is a function of two independent coordinates whose ordinary partial derivatives are simultaneously μ and λz . Indeed,

$$[D_u, D_v] = \mu \lambda \partial_z. \quad (26)$$

When $\mu \lambda \neq 0$, this bracket already rules out such an integrable coordinate interpretation. The same bracket is computed intrinsically in (8); the local model here and the frame there describe one object. Its geometric reading as a contact structure is taken up in Section 9.

Theorem 3.8 (Continuous affine flow). *Let $\mathfrak{E} = (M, g, a; \mu, \lambda)$ be a regular AES with its canonical arithmetic frame. If a unit-speed curve γ has tangent*

$$\dot{\gamma} = \cos \theta X_u + \sin \theta X_v,$$

where θ may vary along the curve, then

$$\frac{d}{ds}(a \circ \gamma) = \mu \cos \theta + \lambda(a \circ \gamma) \sin \theta. \quad (27)$$

For constant θ , its solutions are given by (23).

Proof. By the chain rule and (6),

$$\frac{d}{ds}(a \circ \gamma) = da(\dot{\gamma}) = \cos \theta X_u a + \sin \theta X_v a = \mu \cos \theta + \lambda a \sin \theta.$$

For constant θ , this is the linear equation already integrated in (23). $\square$

Rectification. For $\lambda \neq 0$, define

$$r = \operatorname{arcsinh}\left(\frac{\lambda a}{\mu}\right). \quad (28)$$

The chain rule and (5) give

$$|\nabla r| = |\lambda|. \quad (29)$$

Thus r has constant gradient magnitude. When $\lambda = 0$, the original equation already reduces to $|\nabla a| = |\mu|$, and a/μ is a unit-gradient rectifying function. Rectification of the assignment is possible; rectification of *both* arithmetic directions at once is not, by (26). This asymmetry is the discrete root of the tearing language of Section 11.

The same affine convention fixes the sign of the elementary local order defect. For small increments h and k , let

$$A_h(z) = z + \mu h, \quad M_k(z) = e^{\lambda k} z.$$

Then

$$\begin{aligned} (M_k \circ A_h)(z) - (A_h \circ M_k)(z) &= \mu h(e^{\lambda k} - 1) \\ &= \mu \lambda h k + O(\|(h, k)\|^3). \end{aligned} \quad (30)$$

This is an infinitesimal two-order comparison, not an exact area formula for an arbitrary finite region. The exact finite statement, and its interpretation as a defect measured in the commutativized space, occupy Section 8.

Affine motion inside projective motion. An infinitesimal projective transformation of the line has the Riccati form

$$\dot{z} = \beta + \alpha z + \kappa z^2.$$

The theory developed in this section is the affine slice $\kappa = 0$, with $\beta = \mu \cos \theta$ and $\alpha = \lambda \sin \theta$. Results proved for regular affine AESs do not automatically extend to the quadratic projective flow.

3.7 Model names: E_0 , E_1 , and later examples

The historical drafts of this programme used the labels E_0 and E_1 inconsistently; in some notes the hyperbolic half-plane is called E_0 , in others E_1 . Because later chapters introduce spaces that are *not* smooth, we fix the convention now, once, and record it as a convention rather than a theorem.

Convention 3.9 (Model labels). $\mathfrak{E}_0(\mu, \lambda)$ denotes the basic regular hyperbolic model of Subsection 3.3. For $k \geq 1$, the label $\mathfrak{E}_k$ is used descriptively for an arithmetic expression space whose regular locus is a surface with k declared punctures: points removed from the regular structure and carrying no regular AES data. No classification by k is asserted; the label records the number of declared punctures and nothing else. The reserved historical labels E_k and $E_{\log}$ of earlier drafts are not model classes, and this paper defines neither of them.

Two examples make the convention concrete; both are developed later, and the first one is the only example in this paper whose punctured model is verified in detail.

Example 3.10 (The basic model, $k = 0$). $\mathfrak{E}_0(\mu, \lambda)$ is the upper half-plane with the metric (9) and the assignment $a = -x/y$. It has no puncture, its assignment vanishes along one complete geodesic, and by Theorem 3.4 it is complete with constant curvature. This is the model in which the arithmetic motions of Subsection 3.5 are read as point paths.

Example 3.11 (A one-puncture model, $k = 1$). Let $\mathbb{D} = \{\rho < 1\}$ be the unit disc with the complete Poincaré metric $g_{\mathbb{D}} = 4(d\xi^2 + d\eta^2)/\lambda^2(1 - \rho^2)^2$ and the radial assignment

$$a_{\mathbb{D}}(\xi, \eta) = \begin{cases} \frac{2\mu}{\lambda} \frac{\rho}{1 - \rho^2}, & \rho \neq 0, \\ 0, & \rho = 0. \end{cases}$$

On the punctured disc $\mathbb{D} \setminus \{0\}$ this is a regular AES with $|\nabla a_{\mathbb{D}}|^2 = \mu^2 + \lambda^2 a_{\mathbb{D}}^2$, while at the centre the assignment fails to be C^1 although the metric extends smoothly. The regular locus is therefore $\mathbb{D} \setminus \{0\}$ and the model carries exactly one puncture. Paper I verifies the details of this model as its isolated-zero example; the global invariant $\lambda a_{\mathbb{D}} = \mu \sinh(\lambda s)$ along hyperbolic distance s from the centre shows that the assignment, not the metric, is what fails at the puncture.

Example 3.11 is the first holed arithmetic expression space, and Section 11 returns to it. Its puncture is not a hole in the metric; it is a point at which the regular structure stops being defined, and it is where the arithmetic motions of the punctured part can no longer be extended.

4 Ripple Geometry: The Projective Dual

A point path follows one chosen input through a sequence of motions. The projective operation c/z admits no such motion of the value, and the observation of Subsection 3.5 was that its matrix has a nonzero lower-left entry. There is nevertheless a geometry attached to it, and it is obtained by reversing the direction of the reading: instead of pushing a point forward, pull an entire output condition backward through the same maps. The upper half-plane makes this dual reading elementary, because Möbius maps carry generalized circles to generalized circles.

We call the resulting picture *ripple geometry*: the family of conditions obtained by pulling output conditions back through a projective arithmetic operator. The name is descriptive, and Subsection 4.4 states precisely what it does and does not assert.

4.1 Poles and standard horocycles

Let

$$F(z) = \frac{Az + B}{Cz + D} \quad (31)$$

be a real fractional linear map with

$$\Delta = AD - BC > 0.$$

The positivity convention ensures that F maps the upper half-plane to itself. The projective pole is

$$p_F = F^{-1}(\infty) = \begin{cases} -D/C, & C \neq 0, \\ \infty, & C = 0. \end{cases}$$

For $t > 0$, denote the standard horocycle based at infinity by

$$\mathcal{H}_t = \{w \in \mathbb{H} : \operatorname{Im} w = t\}.$$

Definition 4.1 (Ripple pencil). The *ripple pencil* of F is the one-parameter family

$$\mathcal{R}(F) = \{F^{-1}(\mathcal{H}_t) : t > 0\}.$$

It is a family of horocycles based at the pole p_F . The word “ripple” is a visual name for this standard Möbius pullback family.

Theorem 4.2 (Line-to-circle transition). *Let F be as in (31).*

1. *If $C = 0$, every member of $\mathcal{R}(F)$ is a horizontal line; the pencil is based at ∞ .*
2. *If $C \neq 0$, the member with parameter t is the Euclidean circle*

$$\left(x + \frac{D}{C}\right)^2 + \left(y - \frac{\Delta}{2tC^2}\right)^2 = \left(\frac{\Delta}{2tC^2}\right)^2. \quad (32)$$

All these circles are tangent to the real boundary at the common point $p_F = -D/C$ and are nested as t varies.

Proof. For $z = x + iy$, direct calculation gives the standard identity

$$\operatorname{Im} F(z) = \frac{\Delta y}{|Cz + D|^2}. \quad (33)$$

The inverse image of $\mathcal{H}_t$ is therefore

$$\Delta y = t((Cx + D)^2 + C^2 y^2).$$

If $C = 0$, this reduces to a horizontal line. If $C \neq 0$, divide by tC^2 and complete the square in y :

$$\left(x + \frac{D}{C}\right)^2 + y^2 - \frac{\Delta}{tC^2}y = 0,$$

which is exactly (32). The center lies vertically above $-D/C$ at a height equal to the radius, so the circle is tangent to the boundary there. The radius is proportional to t^{-1} , proving nesting. $\square$

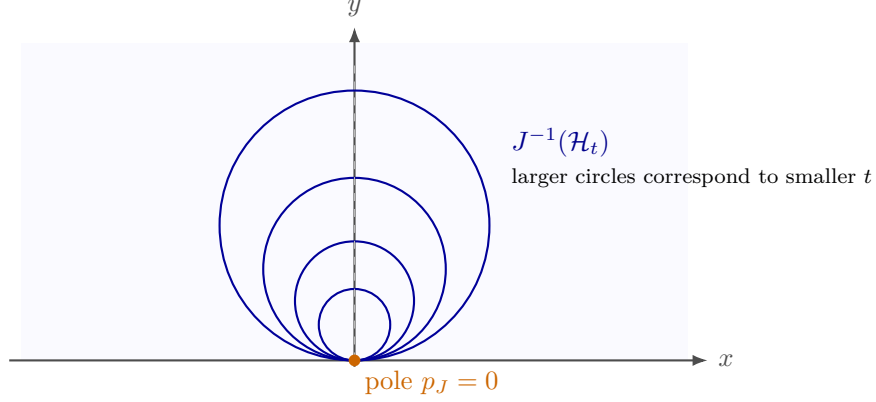


Figure 3: The ripple pencil of inversion. Every circle is a horocycle tangent at the projective pole 0; the affine horizontal-line pencil is recovered when the pole is moved to infinity.

The theorem gives a geometric interpretation of the lower-left matrix entry. The affine condition $C = 0$ places the pole at infinity and degenerates the ripple circles to parallel lines. A projective step with $C \neq 0$ moves the pole to a finite boundary point and curves the same horocyclic family into visible ripples.

Example 4.3 (Arithmetic inversion). For

$$J(z) = -\frac{1}{z}, \quad J \leftrightarrow \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

one has $p_J = 0$ and

$$x^2 + \left(y - \frac{1}{2t}\right)^2 = \left(\frac{1}{2t}\right)^2. \quad (34)$$

Thus inversion turns the horizontal horocycles based at infinity into a nested family of circles tangent at zero. The map exchanges the two ideal points 0 and ∞ .

4.2 Zero–pole pencils in the complex chart

There is a second elementary pencil attached to the same operator. Suppose $A, C \neq 0$ and write its zero and pole as

$$q_F = -\frac{B}{A}, \quad p_F = -\frac{D}{C}.$$

Then the output level set $|F(z)| = r$ is equivalent to

$$|z - q_F| = r \left| \frac{C}{A} \right| |z - p_F|. \quad (35)$$

These are Apollonius circles for the ordered zero–pole pair. The horocyclic ripple pencil and the Apollonius pencil encode different output observables, $\text{Im } F$ and $|F|$, but both show that a projective one-hole operator naturally carries more than a point orbit: it carries a zero, a pole, and a family of level curves organized by them.

4.3 Ripple histories of right-expanded expressions

Let a right-expanded expression determine nondegenerate real Möbius sections $F_1, \dots, F_n$, in chronological order, and write

$$G_k = F_k \circ \dots \circ F_1, \quad G_k(z) = \frac{A_k z + B_k}{C_k z + D_k}.$$

At stage k , the arithmetic value of a chosen input follows the point orbit $G_k(z_0)$. Simultaneously, the output horocycles pull back to

$$\mathcal{R}_k = \mathcal{R}(G_k).$$

The tangency point

$$p_k = G_k^{-1}(\infty) = -\frac{D_k}{C_k}$$

when $C_k \neq 0$ is the current pole of the prefix operator.

Thus a general right-expansion history should not be pictured as one fixed family of concentric circles. Each prefix has its own ripple pencil, and the common tangency point of that pencil may move with k . The appropriate geometric state is

$$(G_k, G_k(z_0), p_k, \mathcal{R}_k) :$$

the operator, the selected point, the pole, and the pulled-back wavefront family.

Proposition 4.4 (Covariant–contravariant duality). *For every prefix operator G_k and every output horocycle $\mathcal{H}_t$,*

$$z_0 \in G_k^{-1}(\mathcal{H}_t) \iff G_k(z_0) \in \mathcal{H}_t.$$

Hence the point path and the ripple history are two readings of the same operator word: one pushes the input forward, the other pulls the output condition backward.

Proof. This is the defining property of inverse image. □

The elementary nature of the proposition is precisely its value. No new geometry is inserted between the two pictures. Their unity is already present in the action of one Möbius matrix.

4.4 The status of ripple geometry

The preceding constructions are elementary and proved. The name “ripple geometry” is not, and it is important to separate the two.

- *What is proved.* A real orientation-preserving Möbius map pulls the standard horocycle $\mathcal{H}_t$ back to a line (when its pole is infinity) or to a circle tangent to the boundary at its pole, and these circles are nested; the pole is the inverse image of infinity; the same prefix matrix determines the point orbit, the pole, the fixed points, and the ripple pencil.
- *What is descriptive.* “Ripple geometry” names the dual reading in which the operator acts on conditions instead of points. It does not assert a new isomorphism class of hyperbolic or projective geometry, a new invariant of arithmetic words, or the existence of a projective arithmetic expression space.

- *What is open.* Whether the ripple family, taken together with its moving pole, is intrinsic to the arithmetic word rather than to a chosen upper-half-plane chart; which path or ripple data descend through relations among history words; and whether a punctured model in the sense of Section 11 carries a ripple structure that survives at the puncture.

The reason for keeping the name is that the dual reading is what makes division homogeneous with the other operations at the level of **conditions**: addition, subtraction, and multiplication by a constant move the value and hence move the contours of the assignment, while second-slot division moves the contours without moving the value affinely. The reason for keeping it provisional is that no intrinsic formulation of that observation is available yet; the current statement of it depends on the standard horocycle $\mathcal{H}_t$ and therefore on one chart.

5 Reciprocals, Poles, Infinity, and Fixed Points

Right-slot division makes three elementary notions meet: reciprocal arithmetic, a pole in an affine chart, and the projective point at infinity. Iteration adds a fourth notion: a finite or infinite expression can be the closure of a recurrence at one of its fixed points. We keep the algebraic fixed-point equation, analytic convergence, and projective continuation distinct.

5.1 Arithmetic reciprocals and projective poles

For $c \neq 0$, the ordinary map

$$J_c(z) = \frac{c}{z}$$

is defined on $K \setminus \{0\}$. Its matrix is

$$J_c \leftrightarrow \begin{pmatrix} 0 & c \\ 1 & 0 \end{pmatrix},$$

so projectively

$$J_c(0) = \infty, \quad J_c(\infty) = 0.$$

This continuation does not make ordinary division by zero admissible.

The reciprocal coordinate centered at a finite point q is

$$w_q(z) = \frac{1}{z - q}. \tag{36}$$

It sends q to infinity and infinity to zero. The rational functions whose only projective pole is q —equivalently, those regular on $\mathbb{P}^1 \setminus \{q\}$ and at the old point infinity—form

$$K[w_q] = K\left[\frac{1}{z - q}\right].$$

The usual polynomial chart $K[z]$ is the corresponding ring when the chosen pole is ∞ . This is a coordinate statement, not a claim that one pole gauge is globally preferred.

By Theorem 4.2, the same q is the common boundary tangency point of the horocycles pulled back through w_q . The denominator zero and the ripple center are therefore two readings of one projective datum.

5.2 A reciprocal from an infinite expansion

Let

$$S_0 = 0, \quad S_{n+1} = 1 + rS_n. \quad (37)$$

Then

$$S_n = 1 + r(1 + r(\cdots(1 + r \cdot 0) \cdots)) = \sum_{k=0}^{n-1} r^k.$$

Scalar multiplication is commutative, so this example does not by itself separate left and right slots; it shows how a reciprocal can arise as recursive closure.

Proposition 5.1 (Finite and infinite geometric expansion). *If $r \neq 1$, then*

$$S_n = \frac{1 - r^n}{1 - r}. \quad (38)$$

If $K = \mathbb{R}$ or $\mathbb{C}$ and $|r| < 1$, then

$$S_n \longrightarrow \frac{1}{1 - r}.$$

Formally, $(1 - r)^{-1} = \sum_{k \geq 0} r^k$ in $K[[r]]$.

Proof. Multiplication of the finite sum by $1 - r$ telescopes to $1 - r^n$. Under $|r| < 1$, $r^n \rightarrow 0$. The formal identity follows coefficient by coefficient. $\square$

The convergence hypothesis is active. For $r = -1$ the sequence oscillates, and for $|r| > 1$ it generally does not converge to the algebraically existing reciprocal.

5.3 Fixed-point closure and escape to infinity

Put

$$T_r(s) = 1 + rs.$$

For $r \neq 1$, its finite fixed point is

$$q_r = \frac{1}{1 - r}. \quad (39)$$

Thus, when $|r| < 1$, the geometric-series truncations converge to the fixed point of their generating update.

For $r \neq 0$, the affine update is invertible and has projective matrix

$$M_r = \begin{pmatrix} r & 1 \\ 0 & 1 \end{pmatrix};$$

it fixes ∞ . The exceptional value $r = 0$ gives the constant update $T_0(s) = 1$, whose matrix is singular and which is not an element of PGL_2 . This degenerate case still has the finite fixed point $q_0 = 1$ but carries no invertible projective dynamics.

As $r \rightarrow 1$, the finite point q_r approaches ∞ in $\mathbb{P}^1$. At $r = 1$, $T_1(s) = s + 1$ has no finite fixed point and has ∞ as its repeated projective fixed point. The pole of $r \mapsto 1/(1 - r)$ is therefore exactly the parameter at which finite fixed-point closure becomes parabolic closure at infinity.

Proposition 5.2 (Affine iteration). *Let $T(z) = az + b$ with $a \neq 0$. If $a \neq 1$, put $q = b/(1 - a)$. Then*

$$T(z) = q + a(z - q), \quad T^n(z) = q + a^n(z - q). \quad (40)$$

Over $\mathbb{R}$ or $\mathbb{C}$, every finite orbit converges to q when $|a| < 1$. If $a = 1$ and $b \neq 0$, then $T^n(z) = z + nb$ and the only projective fixed point is ∞ .

Proof. The relation $(1 - a)q = b$ gives the centered form, and induction gives its n th iterate. $\square$

Algebraic existence, attraction, and projective fixed-point structure are different statements; they coincide only under additional hypotheses.

5.4 Right reciprocal iteration and continued fractions

A stationary right reciprocal update is

$$F(z) = a + \frac{b}{z}, \quad b \neq 0, \quad (41)$$

with matrix

$$M_F = \begin{pmatrix} a & b \\ 1 & 0 \end{pmatrix}.$$

Ordinary iteration is defined only while every intermediate denominator is nonzero. Its finite truncations are generalized continued fractions; the matrix product remains projectively meaningful when a truncation reaches infinity. This is the standard matrix mechanism behind continued fractions [3].

Proposition 5.3 (Reciprocal fixed points). *The finite fixed points of (41) solve*

$$z^2 - az - b = 0. \quad (42)$$

At a nonzero fixed point q ,

$$F'(q) = -\frac{b}{q^2}.$$

Over $\mathbb{C}$, q is locally attracting when $|F'(q)| < 1$ and locally repelling when $|F'(q)| > 1$.

Proof. Multiply $q = a + b/q$ by q and differentiate F . $\square$

Example 5.4 (The golden-ratio expansion). For $F(z) = 1 + 1/z$ and $z_0 = 1$, the orbit is

$$1, \quad 2, \quad \frac{3}{2}, \quad \frac{5}{3}, \quad \frac{8}{5}, \dots$$

The fixed points are

$$\phi = \frac{1 + \sqrt{5}}{2}, \quad -\phi^{-1} = \frac{1 - \sqrt{5}}{2}.$$

Since $|F'(\phi)| = \phi^{-2} < 1$, the positive fixed point is attracting on the relevant positive branch. The infinite continued fraction

$$1 + \frac{1}{1 + \frac{1}{1 + \dots}}$$

denotes ϕ only after this convergence has been established; the fixed-point equation alone does not select an attracting root in every field or domain.

For nonstationary maps

$$F_k(z) = a_k + \frac{b_k}{z}, \quad M_k = \begin{pmatrix} a_k & b_k \\ 1 & 0 \end{pmatrix},$$

the n th truncation is represented by $M_n \cdots M_1$. The nested expression, the recurrence, and matrix multiplication are the same finite calculation.

5.5 General projective fixed points

For

$$F(z) = \frac{Az + B}{Cz + D}, \quad AD - BC \neq 0,$$

a finite fixed point satisfies

$$Cz^2 + (D - A)z - B = 0. \quad (43)$$

In homogeneous coordinates, fixed points are the eigenlines of the representing matrix. Infinity is fixed exactly when $C = 0$. Over an algebraic closure a nonidentity projective map has two fixed points counted with multiplicity; over the base field they may be distinct, repeated, or visible only after a field extension.

The elementary chain is now complete:

$$\text{reciprocal} \longrightarrow \text{pole} \longrightarrow \infty \longrightarrow \text{fixed-point closure} \longrightarrow \text{matrix eigenline}.$$

6 Matrix and Projective Unification

The two geometric pictures now have a common algebraic carrier. A one-hole arithmetic section is represented by a 2×2 matrix; chronological composition is matrix multiplication, point paths are matrix orbits, ripple pencils are inverse images under the same products, poles are inverse images of infinity, and fixed points are eigenlines.

6.1 Elementary arithmetic matrices

Let K be a field. In homogeneous coordinates, write

$$z = [z : 1], \quad \infty = [1 : 0],$$

and let

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

act by

$$M \cdot z = \frac{Az + B}{Cz + D}.$$

Nonzero scalar multiples represent the same element of $\text{PGL}_2(K)$.

The nondegenerate elementary sections have the following representatives.

one-hole section	affine formula	matrix
$C_{+,c}^{(1)} = C_{+,c}^{(2)}$	$z + c$	$\begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix}$
$C_{-,c}^{(1)}$	$z - c$	$\begin{pmatrix} 1 & -c \\ 0 & 1 \end{pmatrix}$
$C_{-,c}^{(2)}$	$c - z$	$\begin{pmatrix} -1 & c \\ 0 & 1 \end{pmatrix}$
$C_{\times,c}^{(1)} = C_{\times,c}^{(2)}$	cz	$\begin{pmatrix} c & 0 \\ 0 & 1 \end{pmatrix}$
$C_{/,c}^{(1)}$	z/c	$\begin{pmatrix} 1 & 0 \\ 0 & c \end{pmatrix}$
$C_{/,c}^{(2)}$	c/z	$\begin{pmatrix} 0 & c \\ 1 & 0 \end{pmatrix}$

Here $c \neq 0$ whenever invertibility requires it. The table has two immediate consequences.

1. Every pure slot-(1) section is affine and fixes ∞ .
2. The projectively new primitive is slot-(2) division. Up to a nonzero scaling, it is inversion.

Ordinary arithmetic and projective continuation remain separate. A matrix in $\mathrm{PGL}_2(K)$ acts everywhere on $\mathbb{P}^1(K)$, while the corresponding ordinary word may be undefined at an intermediate denominator zero.

6.2 Chronological products

Let $F_1, \dots, F_n$ be nondegenerate sections with matrix representatives $M_1, \dots, M_n$, and suppose F_1 is applied first. Define

$$P_k = M_k M_{k-1} \cdots M_1, \quad G_k = F_k \circ \cdots \circ F_1. \quad (44)$$

Then P_k represents G_k .

Theorem 6.1 (One matrix, four readings). *Write*

$$P_k = \begin{pmatrix} A_k & B_k \\ C_k & D_k \end{pmatrix}.$$

The same prefix matrix determines:

1. the point-path state

$$z_k = G_k(z_0) = \frac{A_k z_0 + B_k}{C_k z_0 + D_k};$$

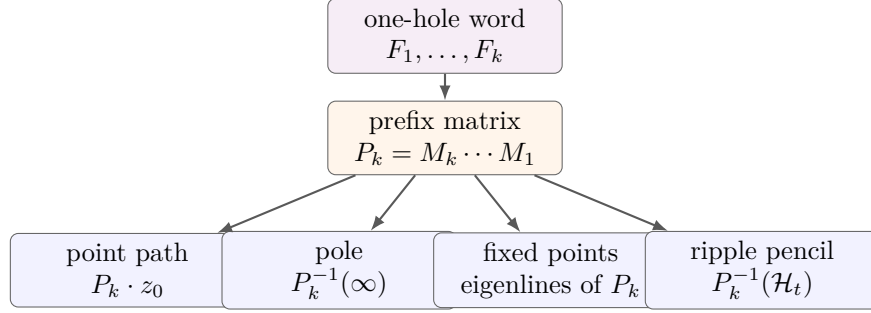


Figure 4: Four geometric readings of the same chronological matrix product.

2. the projective pole

$$p_k = G_k^{-1}(\infty) = -D_k/C_k$$

when $C_k \neq 0$, and $p_k = \infty$ when $C_k = 0$;

3. the fixed-point equation

$$C_k z^2 + (D_k - A_k)z - B_k = 0;$$

4. over $\mathbb{R}$, when a lift has positive determinant, the ripple pencil $G_k^{-1}(\mathcal{H}_t)$ from Theorem 4.2.

Proof. The first formula is the matrix action. The pole solves $C_k z + D_k = 0$. The fixed-point equation comes from $z(C_k z + D_k) = A_k z + B_k$. The final statement is the definition of the ripple pencil together with formula (33). $\square$

Thus *path* and *ripple* are not two unrelated representations. They are the covariant and contravariant geometries of one chronological matrix product.

6.3 The affine chart as the stabilizer of infinity

Let

$$B_\infty = \text{Stab}_{\text{PGL}_2(K)}(\infty).$$

A matrix fixes infinity exactly when $C = 0$. After projective rescaling, every such element has a unique representative

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}, \quad a \in K^\times, \quad b \in K,$$

and acts as $z \mapsto az + b$. Hence

$$B_\infty \cong \text{Aff}(1, K). \tag{45}$$

This identifies the path geometry of Section 3 as a pole-at-infinity gauge. In that gauge:

- the denominator is constant;
- the ripple pencil is a parallel-line pencil;
- fixed-point recursion is affine;
- the lower-left matrix entry stays zero.

Choosing a finite pole q amounts to conjugating this affine subgroup by a projective map sending q to infinity. The coordinate $w_q = 1/(z - q)$ from (36) is the simplest such gauge.

6.4 Generation of the projective group

Introduce

$$T_s(z) = z + s, \quad D_q(z) = qz \quad (q \neq 0), \quad J(z) = -\frac{1}{z}.$$

They arise respectively from addition, multiplication, and second-slot division.

Theorem 6.2 (Elementary projective generation). *For every field K , translations T_s , nonzero scalings D_q , and inversion J generate $\mathrm{PGL}_2(K)$. Equivalently, the projective evaluations of nondegenerate left- and right-slot arithmetic sections have image $\mathrm{PGL}_2(K)$.*

Proof. Let

$$F(z) = \frac{Az + B}{Cz + D}, \quad AD - BC \neq 0.$$

If $C = 0$, then $A, D \neq 0$ and

$$F = T_{B/D} \circ D_{A/D}.$$

If $C \neq 0$, direct calculation gives

$$F = T_{A/C} \circ D_{(AD-BC)/C^2} \circ J \circ T_{D/C}. \quad (46)$$

All scale parameters are nonzero, so every factor is available. These two cases exhaust $\mathrm{PGL}_2(K)$. $\square$

The proof also yields the rank-one Bruhat decomposition

$$\mathrm{PGL}_2(K) = B_\infty \sqcup B_\infty J B_\infty. \quad (47)$$

The first cell consists of operators whose pole is infinity. The second consists of operators with a finite pole in the chosen affine chart. Thus the transition from a point-path grid to a curved ripple pencil is the elementary geometric face of crossing from the affine cell to the projective cell.

6.5 Left and right representations in one projective plane

We can now answer the algebraic comparison of the two combs precisely.

Theorem 6.3 (Projective unification of the two combs). *The left- and right-expanded combs have the same abstract chain of internal dependencies and are exchanged by the opposite-operation involution of Proposition 2.4. Their one-hole operator images have the following relation:*

1. *the pure left-expanded nondegenerate language lies in $B_\infty \cong \mathrm{Aff}(1, K)$;*
2. *the pure right-expanded language already contains translations, nonzero scalings, reflections, and inversion, and therefore generates $\mathrm{PGL}_2(K)$;*
3. *after embedding both languages into $\mathrm{PGL}_2(K)$, their point paths, poles, fixed points, and ripple pencils are all read from the same matrix representation.*

Proof. The dependency and opposition statements were proved in Section 2. The elementary table shows that every slot-(1) generator fixes infinity. The pure slot-(2) language contains T_s because addition is commutative, contains D_q because multiplication is commutative, and contains J through second-slot division. Apply Theorem 6.2. The last assertion is Theorem 6.1. $\square$

The theorem does not say that a left-expanded word and a right-expanded word with the same constants have the same operator. It says that their apparently different elementary geometries live in one projective action and can be compared there without erasing operand-slot information.

7 Affine Cocycles and the Relative Defect

We now restrict attention to the affine motions of Section 3. Every evaluated history then has the form

$$x \mapsto \Phi x + \xi, \quad \Phi \in K^\times, \quad \xi \in K.$$

The accumulated scale Φ and translation ξ retain more of the process than an endpoint value. Their composition law forces two natural translation coordinates. The target-frame coordinate ξ weights each translation by the scales that occur after it; the source-normalized coordinate $\hat{\xi} = \Phi^{-1}\xi$ discounts it by the scales accumulated up to that point. These two coordinates are the discrete data whose commutativized shadow is constructed in Section 8.

Throughout Subsections 7.1–7.4, K is an arbitrary field and every affine multiplier is nonzero. The differential calculation in Subsection 7.3 is over $\mathbb{R}$.

7.1 Discrete affine evaluation

Let

$$f_i(x) = s_i x + t_i, \quad s_i \in K^\times, \quad t_i \in K,$$

and regard $(f_1, \dots, f_n)$ as a chronological history: f_1 acts first. Write

$$f_n \circ \dots \circ f_1(x) = \Phi_n x + \xi_n. \quad (48)$$

For the empty history we set $\Phi_0 = 1$ and $\xi_0 = 0$.

Proposition 7.1 (Affine cocycle formulas). *For a chronological affine history $(f_1, \dots, f_n)$,*

$$\Phi_n = \prod_{i=1}^n s_i, \quad (49)$$

$$\xi_n = \sum_{i=1}^n t_i \prod_{j=i+1}^n s_j. \quad (50)$$

The empty product is 1. Defining the source-normalized translation by

$$\hat{\xi}_n = \Phi_n^{-1} \xi_n, \quad (51)$$

one also has

$$\hat{\xi}_n = \sum_{i=1}^n \frac{t_i}{\prod_{j=1}^i s_j}. \quad (52)$$

Proof. We prove the first two identities simultaneously by induction on n . They hold for $n = 0$. Suppose

$$f_n \circ \dots \circ f_1(x) = \Phi_n x + \xi_n.$$

Postcomposing chronologically by f_{n+1} gives

$$f_{n+1}(\Phi_n x + \xi_n) = s_{n+1} \Phi_n x + (s_{n+1} \xi_n + t_{n+1}).$$

Therefore

$$\Phi_{n+1} = s_{n+1} \Phi_n = \prod_{i=1}^{n+1} s_i.$$

Using the induction hypothesis for ξ_n ,

$$\begin{aligned}\xi_{n+1} &= s_{n+1} \sum_{i=1}^n t_i \prod_{j=i+1}^n s_j + t_{n+1} \\ &= \sum_{i=1}^n t_i \prod_{j=i+1}^{n+1} s_j + t_{n+1} \prod_{j=n+2}^{n+1} s_j,\end{aligned}$$

which is exactly (50) for $n+1$.

Every s_i is nonzero, so Φ_n is invertible. Dividing the i th term of (50) by $\Phi_n = \prod_{j=1}^n s_j$ gives

$$\frac{t_i \prod_{j=i+1}^n s_j}{\prod_{j=1}^n s_j} = \frac{t_i}{\prod_{j=1}^i s_j}.$$

Summation proves (52). □

Formula (50) is a future-weighting rule. A translation t_i is expressed in target coordinates only after every later scale $s_{i+1}, \dots, s_n$ has acted. Formula (52) is a past-normalization rule: the same translation is transported back through the scale accumulated from the source through step i . Neither formula is an additional geometric postulate.

Example 7.2 (Three steps). For

$$f_1(x) = s_1x + t_1, \quad f_2(x) = s_2x + t_2, \quad f_3(x) = s_3x + t_3,$$

direct substitution gives

$$f_3 f_2 f_1(x) = s_3 s_2 s_1 x + s_3 s_2 t_1 + s_3 t_2 + t_3.$$

Here and below juxtaposition of the displayed functions means ordinary functional composition $f_3 \circ f_2 \circ f_1$, not chronological word concatenation. The formula makes the future scales attached to each translation visible.

7.2 Cocycle laws under chronological concatenation

For an affine history γ , write

$$\rho(\gamma)(x) = \Phi_\gamma x + \xi_\gamma, \quad \widehat{\xi}_\gamma = \Phi_\gamma^{-1} \xi_\gamma.$$

Recall that $\gamma\delta$ means first γ and then δ , so $\rho(\gamma\delta) = \rho(\delta)\rho(\gamma)$.

Proposition 7.3 (Chronological affine cocycle law). *For affine histories γ and δ ,*

$$\Phi_{\gamma\delta} = \Phi_\delta \Phi_\gamma, \tag{53}$$

$$\xi_{\gamma\delta} = \Phi_\delta \xi_\gamma + \xi_\delta, \tag{54}$$

$$\widehat{\xi}_{\gamma\delta} = \widehat{\xi}_\gamma + \Phi_\gamma^{-1} \widehat{\xi}_\delta. \tag{55}$$

If γ is invertible, then

$$\Phi_{\gamma^{-1}} = \Phi_\gamma^{-1}, \quad \xi_{\gamma^{-1}} = -\Phi_\gamma^{-1} \xi_\gamma = -\widehat{\xi}_\gamma. \tag{56}$$

Proof. Apply $\rho(\delta)$ to $\rho(\gamma)(x)$:

$$\rho(\delta)(\Phi_\gamma x + \xi_\gamma) = \Phi_\delta \Phi_\gamma x + \Phi_\delta \xi_\gamma + \xi_\delta.$$

This proves (53) and (54). Since K is commutative,

$$\begin{aligned}\widehat{\xi}_{\gamma\delta} &= (\Phi_\delta \Phi_\gamma)^{-1}(\Phi_\delta \xi_\gamma + \xi_\delta) \\ &= \Phi_\gamma^{-1} \xi_\gamma + \Phi_\gamma^{-1} \Phi_\delta^{-1} \xi_\delta,\end{aligned}$$

which is (55). Finally, solving $y = \Phi_\gamma x + \xi_\gamma$ for x yields

$$\rho(\gamma)^{-1}(y) = \Phi_\gamma^{-1} y - \Phi_\gamma^{-1} \xi_\gamma,$$

and hence (56). $\square$

These formulas show why the translation coordinate is a cocycle rather than an additive homomorphism. The semidirect-product action of scale on translation is part of the arithmetic composition law.

7.3 Left and right affine differentials

We now pass to the positive real affine group and write

$$g(M, \xi) = \begin{pmatrix} e^M & \xi \\ 0 & 1 \end{pmatrix}. \quad (57)$$

The symbol M denotes accumulated logarithmic scale; it is distinct from the fixed infinitesimal intensity λ used in later flow equations.

Proposition 7.4 (Left and right affine differentials). *For the matrix g in (57), the translation components of the left and right Maurer–Cartan forms are*

$$(g^{-1} dg)_{\text{tr}} = e^{-M} d\xi, \quad (58)$$

$$(dg g^{-1})_{\text{tr}} = d\xi - \xi dM. \quad (59)$$

If $\widehat{\xi} = e^{-M} \xi$, then

$$d\xi - \xi dM = e^M d\widehat{\xi}. \quad (60)$$

Proof. We have

$$g^{-1} = \begin{pmatrix} e^{-M} & -e^{-M} \xi \\ 0 & 1 \end{pmatrix}, \quad dg = \begin{pmatrix} e^M dM & d\xi \\ 0 & 0 \end{pmatrix}.$$

Multiplication on the two sides gives

$$g^{-1} dg = \begin{pmatrix} dM & e^{-M} d\xi \\ 0 & 0 \end{pmatrix}$$

and

$$dg g^{-1} = \begin{pmatrix} dM & d\xi - \xi dM \\ 0 & 0 \end{pmatrix}.$$

The upper-right entries prove (58) and (59). Differentiating $\widehat{\xi} = e^{-M} \xi$ gives

$$d\widehat{\xi} = e^{-M} (d\xi - \xi dM),$$

which is equivalent to (60). $\square$

The left Maurer–Cartan form $g^{-1}dg$ measures velocity in body or source coordinates; the right form $dg g^{-1}$ measures it in spatial or target coordinates. This left–right group distinction is not the same as operand-slot chirality. Relating a slot choice to left or right group multiplication requires an additional rule specifying how a new context is inserted into a represented history. No such identification is assumed here.

There are likewise two separate coordinate statements. The matrix coordinate ξ is the target-frame translation in $x \mapsto e^M x + \xi$, whereas $\hat{\xi} = e^{-M} \xi$ is its source-normalized value. Equations (58)–(60) relate their differentials but do not make the two coordinates interchangeable.

7.4 Relative affine defect

The affine cocycle permits a comparison of two histories before any global torsion theory is introduced.

Definition 7.5 (Relative affine defect). For two affine histories γ and δ , the *relative affine defect of γ with reference δ* is

$$\mathcal{K}_{\text{aff}}(\gamma, \delta) = \rho(\delta)^{-1} \rho(\gamma) \in \text{Aff}(1, K). \quad (61)$$

The pair is *scale-compatible* when $\Phi_\gamma = \Phi_\delta$.

The order of the arguments in (61) is part of the convention. Interchanging γ and δ inverts the group-valued defect.

Proposition 7.6 (Coordinates of the relative affine defect). *For arbitrary affine histories γ and δ ,*

$$\mathcal{K}_{\text{aff}}(\gamma, \delta)(x) = \frac{\Phi_\gamma}{\Phi_\delta} x + \frac{\xi_\gamma - \xi_\delta}{\Phi_\delta}. \quad (62)$$

If the pair is scale-compatible with common scale Φ , then

$$\nu_x(\gamma) - \nu_x(\delta) = \xi_\gamma - \xi_\delta, \quad (63)$$

$$\mathcal{K}_{\text{aff}}(\gamma, \delta)(x) = x + \frac{\xi_\gamma - \xi_\delta}{\Phi}. \quad (64)$$

In particular, the endpoint difference is then independent of x .

Proof. The inverse of $y \mapsto \Phi_\delta y + \xi_\delta$ is $y \mapsto \Phi_\delta^{-1}(y - \xi_\delta)$. Applying it after $x \mapsto \Phi_\gamma x + \xi_\gamma$ gives (62). If $\Phi_\gamma = \Phi_\delta = \Phi$, direct subtraction of the two endpoint formulas gives (63), and substituting the common scale into (62) gives (64). $\square$

For unequal scales, the endpoint difference contains $(\Phi_\gamma - \Phi_\delta)x$ and depends on the initial value. In that case the group-valued comparison remains defined, but there is no base-point-independent scalar translation defect of the form (63).

Example 7.7 (The elementary affine order defect). Fix $\mu \in K$ and $q \in K^\times$. Let γ first add μ and then scale by q , while δ first scales by q and then adds μ . Then

$$\rho(\gamma)(x) = q(x + \mu) = qx + q\mu, \quad \rho(\delta)(x) = qx + \mu.$$

The histories have the same scale q , and hence

$$\nu_x(\gamma) - \nu_x(\delta) = \mu(q - 1), \quad \mathcal{K}_{\text{aff}}(\gamma, \delta)(x) = x + \mu(1 - q^{-1}). \quad (65)$$

Over the positive real group, writing $q = e^M$ gives the endpoint defect $\mu(e^M - 1)$.

Equation (65) is an open comparison of two chronological histories. It is not a closed commutator-loop holonomy, a comparison with path inverse, or a mirror invariant. Section 8 places compatible two-history defects in the global torsion framework.

7.5 Propagation of a change in the initial accumulator

The accumulated scale also controls sensitivity to the initial value.

Proposition 7.8 (Exact affine perturbation propagation). *Let γ be an affine history with scale Φ_γ . For any $x, \tilde{x} \in K$,*

$$\nu_{\tilde{x}}(\gamma) - \nu_x(\gamma) = \Phi_\gamma(\tilde{x} - x). \quad (66)$$

When $\tilde{x} \neq x$,

$$\frac{\nu_{\tilde{x}}(\gamma) - \nu_x(\gamma)}{\tilde{x} - x} = \Phi_\gamma. \quad (67)$$

Proof. The affine normal form gives

$$\nu_{\tilde{x}}(\gamma) = \Phi_\gamma \tilde{x} + \xi_\gamma, \quad \nu_x(\gamma) = \Phi_\gamma x + \xi_\gamma.$$

Subtracting cancels the translation cocycle and proves (66). Division by the nonzero element $\tilde{x} - x$ proves (67). $\square$

This is an exact discrete identity, not a linear approximation. The continuous arithmetic motion of Section 3 is obtained by differentiating the same semidirect-product action. Supplementary expansions and convention checks appear in Appendix C.

8 Commutativization: The ACS and Arithmetic Tearing

The arithmetic motions of Section 3 do not commute: addition and multiplication transport each other's rulers by the Baumslag–Solitar-type relation (19), and the affine cocycle of Section 7 records the resulting order dependence in two translation coordinates. This section constructs the *commutativization* of that non-commutative motion.

Commutativization here means the following. Keep the total additive charge and the total logarithmic multiplicative charge of a history, and forget the order in which they were accumulated; the charges of an additive step and of a scaling step then commute by construction, and the resulting object is a plane with two coordinates. We call it the *accumulative commutative space*, ACS. The construction has two distinct uses, and they must not be conflated. For a positive add–scale history it records the two total charges; and it supplies a weighted integral along the charge path whose value restores the translation part that the charge endpoint alone forgets. The ACS is therefore a *shadow* of the presented history: it is the commutative quotient object attached to a non-commutative one, not another presentation of the full arithmetic syntax.

The second use is the geometric one, and it is the reason for the name of this chapter. Two histories with the same charge endpoint have charge paths that together bound a region of the ACS, and the exact value of their order defect is a weighted area of that region (Theorem 8.5). In Section 8.6 we read that identity as a statement about the original AES: the arithmetic space carries a defect that cannot be seen in any single motion, and the geometry in which it becomes an area is precisely the commutativization of that space.

Throughout this section an elementary additive step and an elementary scaling step are denoted by

$$A_p(x) = x + p, \quad M_q(x) = e^q x, \quad p, q \in \mathbb{R}.$$

For a history $\gamma = (g_1, \dots, g_n)$, the entries are performed from left to right, so that

$$\nu_x(\gamma) = g_n \circ \dots \circ g_1(x).$$

Negative values of p and q are allowed. Thus the same notation covers inverse translations and inverse positive scalings. In this section an *add-scale history* means a finite word in these two families of generators. Such a word evaluates in $\text{Aff}^+(1, \mathbb{R})$. The ACS construction is attached to this presented word; it is not asserted for negative multipliers or for an arbitrary affine syntax before that syntax has been written in add-scale form.

8.1 The charge path

Definition 8.1 (Accumulative commutative space and charge path). The *accumulative commutative space*, abbreviated ACS, is the oriented plane

$$\mathcal{A} = \mathbb{R}_{(A,M)}^2, \quad \text{with orientation } dA \wedge dM.$$

An additive step A_p has charge increment $(p, 0)$, and a scaling step M_q has charge increment $(0, q)$. Starting at $(A_0, M_0) = (0, 0)$, accumulate these increments in chronological order. The resulting oriented broken path is denoted by C_γ , and its terminal point by

$$(A_\gamma, M_\gamma) = \left(\sum_{g_i=A_{p_i}} p_i, \sum_{g_i=M_{q_i}} q_i \right).$$

The terminal charge is insensitive to order, whereas the broken path generally is not. In particular, two histories may have the same ACS endpoint while inducing different affine maps. The missing information is recovered by the following integral formula.

Proposition 8.2 (Direct ACS evaluation). *For every add-scale history γ and every initial value x ,*

$$\nu_x(\gamma) = e^{M_\gamma} \left(x + \int_{C_\gamma} e^{-M} dA \right). \quad (68)$$

Proof. Let an additive step A_{p_i} occur when the cumulative multiplicative charge is M_{i-1} . Along the corresponding horizontal edge of C_γ ,

$$\int e^{-M} dA = p_i e^{-M_{i-1}}.$$

After that addition, the remaining multiplicative steps have total charge $M_\gamma - M_{i-1}$. Consequently the contribution of p_i to the final value is

$$p_i e^{M_\gamma - M_{i-1}} = e^{M_\gamma} p_i e^{-M_{i-1}}.$$

The initial value contributes $e^{M_\gamma} x$, and vertical ACS edges contribute nothing to the dA -integral. Summing over the additive steps gives (68). The argument uses oriented edge integrals, so it also covers negative increments. $\square$

The kernel e^{-M} is therefore forced by the affine cocycle: it normalizes an additive contribution to the source frame, before the common terminal scale e^{M_γ} returns it to the target frame.

8.2 Compatibility and relative torsion

There are two useful compatibility conditions, and they should not be conflated. Histories γ and δ are *scale-compatible* if

$$M_\gamma = M_\delta.$$

They are *charge-compatible* if, in addition,

$$A_\gamma = A_\delta.$$

Scale compatibility makes their affine maps have the same linear part. Charge compatibility makes their ACS paths have both endpoints in common and hence makes $C_\gamma - C_\delta$ a closed oriented one-chain.

Definition 8.3 (Relative arithmetic torsion). For scale-compatible add-scale histories γ and δ , their *relative arithmetic torsion* is

$$\tau(\gamma, \delta) := \xi_\gamma - \xi_\delta, \quad (69)$$

where ξ_γ and ξ_δ are the target-frame translation terms of the two affine operators. This relative translation defect is the primary torsion invariant used below.

Proposition 8.4. *For every initial value x , relative arithmetic torsion equals the endpoint difference*

$$\tau(\gamma, \delta) = \nu_x(\gamma) - \nu_x(\delta),$$

and is therefore independent of x . If $M_\gamma = M_\delta = M_$, then*

$$\tau(\gamma, \delta) = e^{M_*} \left(\int_{C_\gamma} e^{-M} dA - \int_{C_\delta} e^{-M} dA \right). \quad (70)$$

Proof. Apply Proposition 8.2 to both histories. Their initial terms are respectively $e^{M_\gamma}x$ and $e^{M_\delta}x$, which cancel under scale compatibility. The remaining terms give (70). $\square$

Temporal reversal gives one possible comparison, but it is not built into the definition. The pair (γ, δ) may consist of any two scale-compatible add-scale histories. This extension is important because order defects occur between histories that are neither reversals nor planar mirrors of one another.

8.3 The torsion–Stokes formula

Suppose now that γ and δ are charge-compatible, with common terminal charge (A_*, M_*) . Define the target-normalized one-form

$$\eta_* := e^{M_* - M} dA \quad (71)$$

on the ACS. Relative torsion is then the integral of η_* over the closed one-chain $C_\gamma - C_\delta$. The chosen order of the two paths fixes the sign.

Theorem 8.5 (Weighted torsion–Stokes theorem). *Let γ and δ be charge-compatible add-scale histories with common terminal multiplicative charge M_* . Let $\Sigma_{\gamma, \delta}$ be any oriented piecewise smooth singular two-chain in the ACS satisfying*

$$\partial \Sigma_{\gamma, \delta} = C_\gamma - C_\delta.$$

Then

$$\tau(\gamma, \delta) = \int_{C_\gamma - C_\delta} \eta_* \quad (72)$$

$$= \int_{\Sigma_{\gamma, \delta}} e^{M_* - M} dA \wedge dM. \quad (73)$$

The value of the area integral is independent of the chosen filling chain.

Proof. Equation (70) and charge compatibility give

$$\tau(\gamma, \delta) = \int_{C_\gamma - C_\delta} e^{M_* - M} dA = \int_{C_\gamma - C_\delta} \eta_*.$$

With the orientation $dA \wedge dM$, direct differentiation yields

$$d\eta_* = d(e^{M_* - M}) \wedge dA = -e^{M_* - M} dM \wedge dA = e^{M_* - M} dA \wedge dM.$$

Stokes' theorem for piecewise smooth chains therefore gives

$$\int_{C_\gamma - C_\delta} \eta_* = \int_{\partial \Sigma_{\gamma, \delta}} \eta_* = \int_{\Sigma_{\gamma, \delta}} d\eta_*,$$

which is (73). Such a filling exists because the ACS plane is contractible. Finally, if Σ and Σ' have the same boundary, applying Stokes' theorem to each shows

$$\int_{\Sigma} d\eta_* = \int_{\partial \Sigma} \eta_* = \int_{\partial \Sigma'} \eta_* = \int_{\Sigma'} d\eta_*.$$

This also proves filling independence. The chain formulation remains valid when the broken paths self-intersect or carry signed increments; no informal notion of an enclosed region is required. $\square$

Charge compatibility is needed for the closed-chain and area formulas, but not for Definition 8.3. A merely scale-compatible pair still has a well-defined translation defect, even when the endpoints of its ACS paths have different A -coordinates.

Figure 5 depicts the chain orientation used in Theorem 8.5. It is a schematic of the theorem's data rather than an argument for filling independence.

8.4 Explicit order comparisons

The elementary rectangle. Let

$$\gamma = (A_p, M_q), \quad \delta = (M_q, A_p).$$

The path $C_\gamma - C_\delta$ is the positively oriented boundary of the rectangle $[0, p] \times [0, q]$ when $p, q > 0$. Direct evaluation and the weighted-area formula agree:

$$\begin{aligned} \tau(\gamma, \delta) &= e^q(x + p) - (e^q x + p) \\ &= p(e^q - 1) \\ &= \int_0^p \int_0^q e^{q-M} dM dA. \end{aligned}$$

Thus “add, then multiply” minus “multiply, then add” is positive under the orientation fixed in Definition 8.1. In the flow notation used in the next chapter, $p = \mu h$ and $q = \lambda k$, so the defect is

$$\mu h(e^{\lambda k} - 1).$$

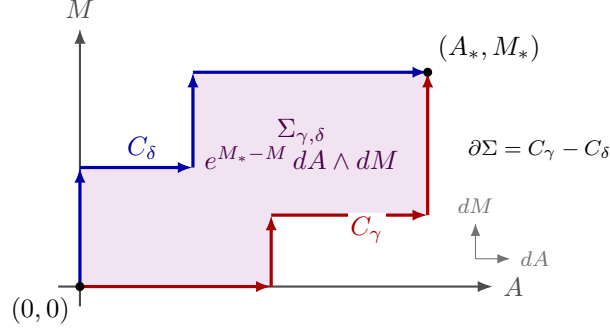


Figure 5: Two charge-compatible ACS histories. Both paths run from $(0,0)$ to the common charge endpoint (A_*, M_*) . Traversing C_γ forward and C_δ backward gives the positive boundary of the displayed filling for the orientation $dA \wedge dM$; the density is weighted by e^{M_*-M} .

A four-step staircase. For

$$\gamma = (A_p, M_r, A_q, M_s), \quad \delta = (M_r, A_p, M_s, A_q),$$

the histories have the same total charge $(p+q, r+s)$, but δ is not the temporal reversal of γ . Their evaluations are

$$\begin{aligned} \nu_x(\gamma) &= e^{r+s}x + pe^{r+s} + qe^s, \\ \nu_x(\delta) &= e^{r+s}x + pe^s + q. \end{aligned}$$

Consequently

$$\tau(\gamma, \delta) = pe^s(e^r - 1) + q(e^s - 1). \quad (74)$$

For positive p, q, r, s , both terms are positive weighted strips in an ACS filling. Formula (74) also shows how a later scale changes the weight of an earlier commutation defect.

Equal charges without reversal. The pair

$$(A_p, M_r, A_q), \quad (A_q, M_r, A_p)$$

has common charge $(p+q, r)$, and its relative torsion is

$$(p-q)(e^r - 1).$$

It may be positive, negative, or zero. In particular, equal charge does not determine torsion, while zero torsion does not imply equality of marked histories.

8.5 What the ACS forgets

At the formal level, the map

$$\gamma \mapsto (A_\gamma, M_\gamma)$$

is an order-forgetting charge homomorphism from concatenated marked histories to the additive group $\mathbb{R}^2$. It is therefore analogous to an abelianization map. This statement concerns the formal charge language only. It does not identify the symbolic history space with the evaluated affine group, and it does not assert that either object is a free group without an additional choice of formal generators and relations.

The content of Theorem 8.5 is precisely what survives this loss of order: the endpoint charges specify a common frame, while the weighted filling measures the translation defect between two histories. Section 9 retains the current value as a third coordinate and identifies the infinitesimal form of the same defect with contact curvature.

8.6 Tearing

We can now name the phenomenon that the three preceding chapters have been describing from three sides. The name is new here, the mathematics is not: it is the content of Definition 8.3 and Theorem 8.5 read geometrically.

Definition 8.6 (Tearing of arithmetic motion). Let γ and δ be two arithmetic histories of a regular AES that are scale-compatible in the sense of Section 8, and let $\tau(\gamma, \delta)$ be their relative torsion. The pair is said to *tear* the AES by $\tau(\gamma, \delta)$: the two histories induce the same linear part, hence agree on the multiplicative content of the motion, yet their additive content differs, and the difference is exactly the weighted ACS filling between their charge paths.

Three properties justify treating tearing as a geometric notion rather than a bookkeeping defect.

1. *It exists only relative to the commutativization.* In the AES itself, γ and δ are simply two motions; there is no region bounded by them, because the AES is where the two rulers fail to commute. The region appears only after passing to the ACS, and the weight of the filling is forced by the evaluation formula (68).
2. *It is measured, not merely detected.* The value $\tau(\gamma, \delta)$ is an exact weighted area (73), and for the elementary rectangle it is $\mu h(e^{\lambda k} - 1)$, which is the discrete Baumslag–Solitar transport of (19) read as a defect.
3. *It is infinitesimally rigid.* Its density is the constant $\mu\lambda$ of (30), which is also the bracket coefficient of the arithmetic frame (8). Section 9 shows that this coefficient is the curvature of a contact structure, hence that tearing cannot be removed by any local recalibration of the two arithmetic directions.

The word *tearing* is chosen because the two-dimensional picture is that of a surface cut along the two motions: an additive edge and a multiplicative edge that close up on the level of total charge but not on the level of motion. The standard description of a rip in a material is also a pair of paths that agree far away and separate inside; the arithmetic version of that picture is Definition 8.6. As with ripple geometry, the terminology is descriptive and asserts nothing beyond the displayed identities; what it adds is a single name for the phenomenon whose exact measure is the ACS area, whose infinitesimal density is $\mu\lambda$, and whose carrier in a punctured model is the subject of Section 11.

9 The Contact Structure of Tearing

Section 8 measured tearing by a weighted area in the commutativization of the arithmetic motions. That measurement is exact at every finite scale, but it is stated in a space that has forgotten the current value. This section recovers the local picture: a three-dimensional state space that retains the value and distinguishes the two infinitesimal arithmetic directions, carrying an Ehresmann connection over the two-dimensional direction plane and, in the nondegenerate case, a contact form. Tearing then appears a third time, with the same density: as the curvature of that connection.

Fix real constants μ and λ , and let

$$\mathcal{C} = \mathbb{R}_{(u,v,a)}^3, \quad \pi : \mathcal{C} \longrightarrow \mathbb{R}_{(u,v)}^2.$$

Here a is the current assignment value, while u and v parametrize additive and multiplicative motion. We orient $\mathcal{C}$ by

$$du \wedge da \wedge dv.$$

9.1 The arithmetic contact form

Define

$$\alpha := da - \mu du - \lambda a dv. \quad (75)$$

The equation $\alpha = 0$ says that a lifted base motion obeys the affine propagation law.

Proposition 9.1 (Contact nondegeneracy). *With the orientation $du \wedge da \wedge dv$,*

$$\alpha \wedge d\alpha = \mu\lambda du \wedge da \wedge dv. \quad (76)$$

Hence α is a contact form if and only if $\mu\lambda \neq 0$.

Proof. Since μ and λ are constant,

$$d\alpha = -\lambda da \wedge dv.$$

It follows that

$$\begin{aligned} \alpha \wedge d\alpha &= (da - \mu du - \lambda a dv) \wedge (-\lambda da \wedge dv) \\ &= \mu\lambda du \wedge da \wedge dv. \end{aligned}$$

This three-form is nowhere zero exactly when $\mu\lambda \neq 0$. □

The contact-isomorphism class is the standard local one; the arithmetic content lies in the coordinates and in the preferred horizontal frame introduced next. Even in a degenerate case, $\ker \alpha$ remains a well-defined horizontal distribution, although it is then not contact.

9.2 Horizontal lifts and the affine flow

Set

$$D_u := \partial_u + \mu \partial_a, \quad D_v := \partial_v + \lambda a \partial_a. \quad (77)$$

Then

$$\alpha(D_u) = \alpha(D_v) = 0,$$

and the two fields project respectively to ∂_u and ∂_v . Thus

$$\mathcal{D} := \ker \alpha = \text{span}\{D_u, D_v\}$$

is an Ehresmann connection for π . Its additive and multiplicative flows are

$$\Phi_u^h(u, v, a) = (u + h, v, a + \mu h), \quad (78)$$

$$\Phi_v^k(u, v, a) = (u, v + k, e^{\lambda k} a). \quad (79)$$

For a direction angle θ , define

$$D_\theta = \cos \theta D_u + \sin \theta D_v.$$

If $(u(s), v(s), a(s))$ is tangent to D_θ , then

$$\frac{du}{ds} = \cos \theta, \quad \frac{dv}{ds} = \sin \theta,$$

and

$$\frac{da}{ds} = \mu \cos \theta + \lambda a \sin \theta. \quad (80)$$

Thus the horizontal lift reproduces the continuous affine flow rather than postulating a separate dynamics.

9.3 Horizontal curvature

Theorem 9.2 (Contact curvature). *The preferred horizontal fields satisfy*

$$[D_u, D_v] = \mu\lambda\partial_a. \quad (81)$$

In particular, when $\mu\lambda \neq 0$, the horizontal distribution is not integrable. Its bracket defect is vertical and has constant coefficient $\mu\lambda$ in the arithmetic frame.

Proof. Using (77),

$$\begin{aligned} [D_u, D_v] &= [\partial_u + \mu\partial_a, \partial_v + \lambda a\partial_a] \\ &= \mu\partial_a(\lambda a)\partial_a - \lambda a\partial_a(\mu)\partial_a \\ &= \mu\lambda\partial_a. \end{aligned}$$

All remaining coordinate brackets vanish. Since ∂_a is vertical and is not contained in $\ker \alpha$, the bracket is the vertical failure of horizontal closure. $\square$

The sign in (81) agrees with the chronological convention of the preceding chapter. Equivalently,

$$\alpha([D_u, D_v]) = \mu\lambda, \quad d\alpha(D_u, D_v) = -\mu\lambda.$$

The two formulas have opposite signs by the identity $d\alpha(X, Y) = -\alpha([X, Y])$ for horizontal fields X, Y .

9.4 Open defects and closed horizontal loops

The bracket is infinitesimal. At finite scale there are two related, but different, quantities. First compare the two open lifts from (u, v) to $(u + h, v + k)$:

$$\begin{aligned} \Phi_v^k \circ \Phi_u^h(u, v, a) &= (u + h, v + k, e^{\lambda k}(a + \mu h)), \\ \Phi_u^h \circ \Phi_v^k(u, v, a) &= (u + h, v + k, e^{\lambda k}a + \mu h). \end{aligned}$$

Their assignment difference, with “add then multiply” taken first, is

$$T_{\text{open}}(h, k) = \mu h(e^{\lambda k} - 1). \quad (82)$$

This is exactly the elementary ACS rectangle torsion with charge increments $(\mu h, \lambda k)$.

Now perform the chronological four-step history

$$A_h, \quad M_k, \quad A_{-h}, \quad M_{-k},$$

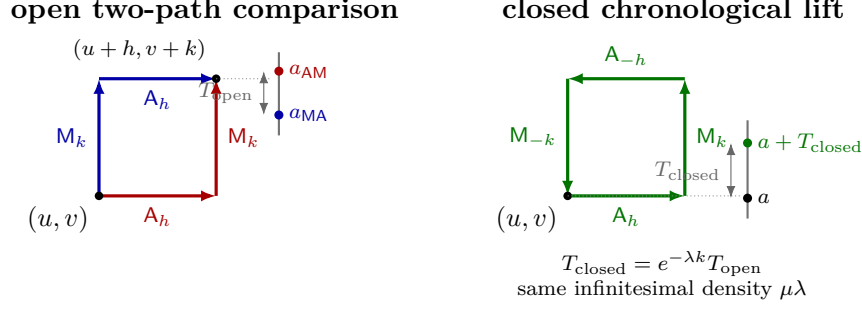


Figure 6: Open defect and closed holonomy in the contact model. The two open horizontal lifts have the same base endpoint but different assignment values. The four-step closed base loop returns to its starting base point with a raw vertical drift. These finite quantities differ by the displayed scale factor; converting the closed drift to the target frame gives the open defect, while their infinitesimal densities agree as stated in the text.

where A_h denotes the flow Φ_u^h and M_k denotes Φ_v^k . The base point returns to (u, v) , and direct substitution gives

$$\begin{aligned} a &\mapsto a + \mu h \mapsto e^{\lambda k}(a + \mu h) \\ &\mapsto e^{\lambda k}(a + \mu h) - \mu h \mapsto a + \mu h(1 - e^{-\lambda k}). \end{aligned}$$

The closed-loop vertical drift is therefore

$$T_{\text{closed}}(h, k) = \mu h(1 - e^{-\lambda k}) = e^{-\lambda k} T_{\text{open}}(h, k). \quad (83)$$

Thus the finite open comparison and the finite closed holonomy are not equal. Here T_{closed} is the source-normalized relative translation. Converting it to the target frame gives $e^{\lambda k} T_{\text{closed}} = T_{\text{open}}$ exactly.

Figure 6 keeps the two finite constructions visually separate. The square is drawn in the (u, v) base; the short vertical segments represent assignment fibers, not additional base edges.

9.5 The scalar horizontal differential

For a smooth scalar field $F \in C^\infty(\mathcal{C})$, define

$$\delta_H F := dF - (\partial_a F) \alpha = (D_u F) du + (D_v F) dv. \quad (84)$$

This is the ordinary differential restricted to horizontal directions and written on the base coframe. In particular,

$$\delta_H a = \mu du + \lambda a dv.$$

For scalar fields only, define the antisymmetrized second horizontal derivative by

$$\mathcal{R}_H(F) := (D_u D_v F - D_v D_u F) du \wedge dv. \quad (85)$$

This notation deliberately avoids presupposing an extension of δ_H to a graded complex.

Proposition 9.3 (Curvature on scalar fields). *For every $F \in C^\infty(\mathcal{C})$,*

$$\mathcal{R}_H(F) = \mu \lambda (\partial_a F) du \wedge dv. \quad (86)$$

In particular,

$$\mathcal{R}_H(a) = \mu \lambda du \wedge dv.$$

Proof. By definition and Theorem 9.2,

$$\mathcal{R}_H(F) = [D_u, D_v](F) du \wedge dv = \mu \lambda (\partial_a F) du \wedge dv.$$

Taking $F = a$ proves the final formula. $\square$

One may use $\delta_H^2 F$ as shorthand for $\mathcal{R}_H(F)$ after declaring this antisymmetrization convention. It does not mean that δ_H has been defined on horizontal forms, nor does it assert the existence of a nilpotent differential complex. No such graded calculus is required for the foundational result.

9.6 Three levels of the same order defect

Proposition 9.4 (Finite and infinitesimal comparison). *For the elementary rectangle with additive side μh and logarithmic multiplicative side λk , the following statements hold.*

1. *The open affine defect equals the ACS weighted area exactly:*

$$T_{\text{open}}(h, k) = \int_{[0, \mu h] \times [0, \lambda k]} e^{\lambda k - M} dA \wedge dM = \mu h (e^{\lambda k} - 1),$$

with oriented-chain interpretation when a side is negative.

2. *The closed chronological lift has exact drift*

$$T_{\text{closed}}(h, k) = e^{-\lambda k} T_{\text{open}}(h, k).$$

Thus T_{closed} is the source-normalized relative translation; its target-frame conversion agrees with the open defect, but the raw finite values generally differ.

3. *Both defects have the common infinitesimal density*

$$\lim_{(h, k) \rightarrow (0, 0)} \frac{T_{\text{open}}(h, k)}{hk} = \lim_{(h, k) \rightarrow (0, 0)} \frac{T_{\text{closed}}(h, k)}{hk} = \mu \lambda,$$

where the quotients are taken for $hk \neq 0$. This is the coefficient of $[D_u, D_v]$ and of $\mathcal{R}_H(a)$.

Proof. The first identity is Theorem 8.5 applied to the elementary rectangle, and its integral evaluates to

$$\mu h \int_0^{\lambda k} e^{\lambda k - M} dM = \mu h (e^{\lambda k} - 1).$$

The second identity is (83). Finally,

$$\frac{e^{\lambda k} - 1}{k} \longrightarrow \lambda, \quad \frac{1 - e^{-\lambda k}}{k} \longrightarrow \lambda,$$

which proves the two limits. The curvature identifications follow from Theorem 9.2 and Proposition 9.3. $\square$

The synthesis has three layers, not one finite equality: exact target-frame open affine torsion is measured by ACS weighted area; source-normalized closed horizontal holonomy differs by an exact frame-conversion factor; and contact curvature is their common infinitesimal limit. The contact distribution alone supplies neither a preferred complex structure nor a full analytic function theory. The present paper uses only the horizontal connection, its scalar differential, and its curvature.

10 Beyond Linear Language

The picture developed so far is deliberately written in the language of surfaces, metrics, vector fields, and contact forms. That language is convenient, and it is also, in a precise sense, too small. This section records two elementary obstructions that explain why we do not take linear language (manifolds, tangent spaces, Lie groups, and character theory built on modules) as the foundation of AEG, and it records the alternative that a companion programme, process geometry, has begun to develop. The obstructions are theorems; the alternative is a structural proposal, and the two are kept apart throughout.

10.1 Two obstructions to linear language

The first obstruction concerns the Lie-algebraic reading of the arithmetic tower. Addition and multiplication are the first two ranks of that tower: an additive step moves the assignment by a constant, a multiplicative step scales it. On the positive assignment line $a > 0$ these two kinds of motion are generated by

$$A = \partial_a, \quad M = a\partial_a.$$

Together with the third rank, multiplication viewed as an endomorphism of addition, they generate more than a three-dimensional Lie algebra.

Proposition 10.1 (The arithmetic generators do not close). *Let*

$$A = \partial_a, \quad M = a\partial_a, \quad P = a \log a \partial_a$$

be defined for $a > 0$. Then

$$[A, M] = A, \quad [M, P] = M, \quad [A, P] = (1 + \log a)\partial_a.$$

The last field does not lie in the real span of A, M, P , and the Lie algebra generated by A, M, P is infinite-dimensional.

Proof. The three brackets are direct: for any smooth f ,

$$[A, M]f = \partial_a(af') - a\partial_a f' = f',$$

$$[M, P]f = a\partial_a(a \log a f') - a \log a \partial_a(af') = af',$$

$$[A, P]f = \partial_a(a \log a f') - a \log a \partial_a f' = (1 + \log a)f'.$$

For the last claim, suppose first that $(1 + \log a)\partial_a = \alpha A + \beta M + \gamma P$ for constants α, β, γ . Equivalently $1 + \log a = \alpha + \beta a + \gamma a \log a$ for all $a > 0$. Letting $a \rightarrow 0^+$ makes the left-hand side tend to $-\infty$ while the right-hand side tends to α , a contradiction. Hence the generated Lie algebra is strictly larger than $\text{span}\{A, M, P\}$.

More is true. Since $[A, a^k \log a \partial_a] = (ka^{k-1} \log a + a^{k-1})\partial_a$ for every $k \geq 1$, the derived algebra contains $a^k \log a \partial_a$ for every $k \geq 1$. The coefficient functions $\{1, a^k, a^k \log a : k \geq 1\}$ are linearly independent over $\mathbb{R}$ on $a > 0$, so these fields span an infinite-dimensional Lie algebra. A fortiori so does the algebra generated by A, M, P . $\square$

The consequence for the present paper is narrow but decisive. The two arithmetic motions of a regular AES are a frame on a surface, and in the basic hyperbolic model they even exponentiate into the positive affine group. That is exactly why E_0 is a good first model. But as soon as the

next rank of the arithmetic tower is admitted, no finite-dimensional Lie group can carry all the motions: the tower generates an infinite-dimensional algebra, and modelling it by three coordinates on a finite-dimensional manifold is not an option to be refused on grounds of taste but a structure that does not exist. The same computation, in the calibration of the companion programme, is recorded as the first strict reason not to model the arithmetic tower as coordinates on an ordinary finite-dimensional manifold [6].

The second obstruction concerns what a *commutative* description can see. It is the obstruction that makes the tearing of Section 8 irreducible rather than an artefact of the presentation.

Proposition 10.2 (Commutativization cannot see tearing). *Let χ be any character of the charge group $\mathbb{R}^2$ of the ACS, and let γ, δ be charge-compatible add-scale histories with common charge (A_*, M_*) . Then*

$$\chi(A_\gamma, M_\gamma) = \chi(A_\delta, M_\delta),$$

while their evaluation defect is the relative torsion

$$\nu_x(\gamma) - \nu_x(\delta) = \tau(\gamma, \delta),$$

which is independent of x and need not vanish. Consequently no description of arithmetic motion by characters of its commutativization can detect tearing.

Proof. The first display is immediate from charge compatibility. The second is Proposition 8.4. Nonvanishing is witnessed by the elementary rectangle: for $\gamma = (A_p, M_q)$ and $\delta = (M_q, A_p)$ one has $\tau(\gamma, \delta) = p(e^q - 1)$, which is nonzero whenever $p \neq 0$ and $q \neq 0$. $\square$

Proposition 10.2 is the precise sense in which the linear language fails here, and it is worth stating without embellishment. Characters, spectra, and module-theoretic decompositions are tools for abelian structures; the ACS is the abelian structure naturally attached to arithmetic motion; and the whole content of the order defect is *outside* what that abelian structure can resolve. A description of AEG that begins from characters therefore begins by discarding the phenomenon the theory is about. This is the technical reason for the methodological position taken in this paper: linear language is used where it is proved to apply (Sections 3–9), and it is not used as the ontology of the subject.

10.2 What process geometry proposes instead

The companion programme process geometry develops a vocabulary for exactly this situation, and the vocabulary is quotable [6]. Its central methodological statement is that geometry is not supplied in advance but generated by the process and by what its observers can distinguish; and that no familiar structure may be imported into a *native* description merely because it is familiar:

- geometry enters "because process generation and observer-relative distinguishability acquire stable relational, local, and compositional structure—not because a manifold, metric, coordinate system, or arithmetic model was supplied in advance";
- "no level should be imported merely because the next level is mathematically familiar";
- the public vocabulary deliberately contains *no* group or Lie-group hierarchy, *no* general representation hierarchy, and *no* spectrum or eigenmode ontology; those "may become justified later, but only after an executable vignette requires it";

- where a classical mechanism (power series, ordinary polynomial algebra, matrix linearization, Fourier or spectral methods, generic computer algebra) is used, it must be declared as such and accompanied by a *lowering witness* scoped to the exact task and lane, rather than silently acquiring the status of native structure.

Two features of that programme are relevant to AEG, and only two are claimed here.

First, it introduces structure without presupposing a linear carrier. A process is a generation rule; a history is what that rule produces; a task selects which distinctions matter; and the resulting quotient, when it is stable, may be *objectified* into a new primitive of the next rank. The vocabulary is deliberately non-linear: it speaks of presentations, semantic compression, rank raising, and rank lowering, and its stated obligations are compositional (a higher-rank composition must remain interpretable at the lower rank) rather than algebraic in the module-theoretic sense.

Second, it records an obstruction of the same kind as Proposition 10.2. Its finite-family layer introduces families with a parameter composition law, together with one-dimensional response objects, and then states plainly that such one-dimensional scalar response is *not* sufficient to encode the additive–multiplicative interaction while retaining a nontrivial translation response. The layer stops at that obstruction instead of enlarging the ambient linear structure; the accompanying declaration says that the object "does not require inverses, topology, Haar measure, Lie algebra, commutativity, or even a group structure". This is the same lesson as Proposition 10.2, arrived at from the side of computation.

10.3 Exact calibrations that meet this paper's laws

The proposals above would be of little use if they did not meet the identities proved here. They do, and in one case the match is literal.

1. *The discrete transport law.* In the calibration of the companion programme, the objectified multiplication acts on translation by

$$D_k T_a = T_{ka} D_k,$$

and the mixed language generated by both is the positive affine monoid $\mathbb{Z} \rtimes \mathbb{N}_{>0}$. This is exactly the relation (19) of Section 3, read in the equivalent direction convention:

$$Y_k X_s Y_k^{-1} = X_{ks},$$

with $D_k = Y_k$ and $T_a = X_a$. The two programmes therefore meet at the same discrete law, one derived from the arithmetic grid and the other from a rank-lowering obligation.

2. *The infinitesimal transport law.* In the continuous form the same structure appears as $S_s T_t = T_{est} S_s$ together with the algebra relation $[A, M] = A$ for $A = \partial_a$ and $M = \partial_v + a\partial_a$. This is the affine structure equation already used in Section 3. It must not be confused with the horizontal bracket (26): the latter is the curvature of the horizontal distribution in the three-dimensional contact model of Section 9, whereas $[A, M] = A$ is a bracket on the base algebra of the two arithmetic motions. The two are related — both have the same coefficient $\mu\lambda$ in the arithmetic frame — but they are different objects, and the identification is not made here.
3. *Non-integrability in a second calibration.* The companion programme also records the exact local obstruction

$$\vartheta = dz - x dy, \quad X = \partial_x, \quad Y = \partial_y + x\partial_z, \quad [X, Y] = \partial_z, \quad \vartheta([X, Y]) = 1.$$

Since $d\vartheta = -dx \wedge dy$, one has $\vartheta \wedge d\vartheta = \pm dx \wedge dy \wedge dz \neq 0$: ϑ is a contact form whose kernel does not close, exactly as for the arithmetic contact form (75). The two forms are not the same object — in α the multiplier of the multiplicative direction is the assignment a itself, whereas in ϑ the coefficient x is an independent coordinate — but the non-integrability pattern is identical: a Pfaffian form on a three-dimensional space whose kernel is spanned by two fields that fail to close, with a nonzero constant bracket component. The same pattern therefore appears in a calibration where no arithmetic assignment is present at all, which is evidence that the phenomenon belongs to the two directions rather than to the assignment.

These three items are exact computations, not proposals. Their status is nevertheless local to their own calibration: they are recorded in a research repository whose own maturity labels are, for the items cited, of the kind "precise conjecture" and "research calibration", and this paper does not promote them.

10.4 Status and non-claims

For the avoidance of doubt, this section claims the following and nothing more.

- Proposition 10.1 and Proposition 10.2 are proved above, by elementary computation, in this paper.
- The vocabulary of Subsection 10.2 is a *structural proposal*: it is a way of organizing the construction of non-linear descriptions, imported from a companion programme and used here as a language, not as a theorem about AEG.
- This section does not claim that manifolds, metrics, Lie groups, modules, or characters are illegitimate in AEG. Sections 3, 8, and 9 use all of them, and use them with proofs. What is refused is their status as ontology: no linear structure is imported into a foundational claim without a derivation of its transport at the arithmetic rank in question.
- This section does not claim any universality of arithmetic-generated geometry, and it does not import the companion programme's own open conjectures. Where that programme states that a claim is not established, the same claim is not established here.
- No efficiency, complexity, or approximation statement is made. The two propositions above are structural obstructions, and an obstruction is not a lower bound: no conclusion about computational cost follows from them, in either direction.

11 Holed Arithmetic Expression Spaces and the Language of Tearing

The models of Section 3 are complete and unpunctured: the basic hyperbolic space has no point at which the arithmetic structure fails. This final chapter records the opposite situation, which arose in the most recent exploratory work of the programme, and the vocabulary that has been proposed for it. The mathematical content of the chapter is deliberately modest: one definition, one exact local computation, and a register of six computed witnesses, each with its own status. The interpretive reading of those witnesses — the language of tearing — is presented as a structural proposal, and Subsection 11.5 lists what is not established.

11.1 Punctured arithmetic expression spaces

Definition 11.1 (Punctured arithmetic expression space). Let $\mathfrak{E} = (M, g, a; \mu, \lambda)$ be a regular AES in the sense of Definition 3.1, and let $S \subset M$ be a finite set of points such that

1. $(M \setminus S, g|_{M \setminus S}, a|_{M \setminus S}; \mu, \lambda)$ is a regular AES, and
2. no neighbourhood of any $p \in S$ admits smooth extensions of $g|_{M \setminus S}$ and $a|_{M \setminus S}$ making it a regular AES.

Then $\mathfrak{E}_S = (M \setminus S, g, a; \mu, \lambda)$ together with the declared set S is called a *punctured arithmetic expression space*, and the points of S are its *punctures*. When $|S| = k$ we write $\mathfrak{E}_k$ for it, in the descriptive sense of Convention 3.9.

The second condition is the one that matters: a puncture is not any point that has been deleted, but a point at which the regular structure genuinely cannot be continued. This is the finite-point case of the singular arithmetic expression space of Paper I; the general singular theory, the classification of singular germs, and everything concerning tubes, braids, and knot invariants belong to Paper I and Paper III. The notion is specialized here for one reason only: the arithmetic motions of Subsection 3.5 are defined on the punctured surface, and the question of what happens to them at a puncture is a question about AES.

The first example is the one-puncture model of Example 3.11: the unit disc with the complete Poincaré metric and the radial assignment $a_{\mathbb{D}} = \frac{2\mu}{\lambda} \rho / (1 - \rho^2)$, whose metric extends smoothly across the centre while the assignment does not. There $S = \{0\}$ and the regular locus is $\mathbb{D} \setminus \{0\}$.

11.2 The local shape of a puncture

A puncture is not merely a missing point; at least in the example that has been computed in detail, it carries a definite local shape. The following statement is exact.

Proposition 11.2 (Tangent cone of the four limiting circles). *Fix $\rho > 0$ and consider the four circles through the origin*

$$C_{\pm} : x^2 + y^2 \mp 2\rho y = 0, \quad D_{\pm} : x^2 + y^2 \mp 2\rho x = 0.$$

Write $S_2 = x^2 + y^2$ and let

$$P = (S_2^2 - 4\rho^2 y^2)(S_2^2 - 4\rho^2 x^2)$$

be the defining polynomial of their union. Then

$$P = S_2^4 - 4\rho^2 S_2^3 + 16\rho^4 x^2 y^2,$$

so the lowest-degree homogeneous part of P is $16\rho^4 x^2 y^2$. Consequently the tangent cone of the union at the origin is

$$(xy)^2 = 0,$$

that is, the two coordinate axes, each with multiplicity two.

Proof. The four circles are $S_2 = \pm 2\rho y$ and $S_2 = \pm 2\rho x$, whose defining polynomials are $S_2^2 - 4\rho^2 y^2$ and $S_2^2 - 4\rho^2 x^2$ respectively; the product is P . Expanding,

$$(S_2^2 - 4\rho^2 y^2)(S_2^2 - 4\rho^2 x^2) = S_2^4 - 4\rho^2 S_2^2(x^2 + y^2) + 16\rho^4 x^2 y^2 = S_2^4 - 4\rho^2 S_2^3 + 16\rho^4 x^2 y^2,$$

since $S_2^2(x^2 + y^2) = S_2^3$. The terms have degrees eight, six, and four, so the lowest-degree part is $16\rho^4 x^2 y^2$, whose zero set is $xy = 0$ counted twice. $\square$

Two consequences are used below. First, the two circles of the same family are tangent at the origin, so the origin is a degenerate point of each family, while circles of different families cross there with tangent lines $x = 0$ and $y = 0$; the tangent cone is the union of those two lines. Second, the tangent cone is *reducible with multiplicity*: it is the square of a product of two distinct linear forms, which is exactly the algebraic shape of a degenerate puncture in this family. The computation is recorded in the exploration register of the programme [5]; the identity above is verified here directly.

11.3 Tearing at a puncture

The tearing of Definition 8.6 is a defect between two motions of a space. On a punctured space a second phenomenon appears: the motions themselves need not be extendable across the puncture, so the comparison of two motions can require passing around it.

Concretely, let $\mathfrak{E}_S$ be a punctured AES and let $p \in S$. A motion of the punctured surface is a path in $M \setminus S$, and two motions that approach p from different directions need not have limits related by any regular motion of $\mathfrak{E}_S$: the assignment and the metric are defined only away from p . Two consequences follow, and both are already visible in the two computed structures of this chapter.

1. *Comparison may require a detour.* In the frame calculation of Subsection 11.4, one period of the phase returns to its starting phase while shifting the label read by the observer. The phase loop is closed in the phase coordinate and open in the label coordinate; the defect is only visible to an observer that records both.
2. *Comparison may require a cut.* In the gluing calculation of the same subsection, the invariants of the whole family of cuts are exhausted by a permutation, and the three-holed object is obtained only after removing two further points. The "holes" are therefore not produced by the gluing; they are the places where the gluing's own data must be checked.

Both statements are deliberately weak. They say where the defect is carried, not what it is equal to. The measured value of a tearing is the ACS area of Theorem 8.5, which is a statement about the unpunctured motion; the puncture is where that measurement cannot be performed locally, and hence where it must be performed either in the commutativization or by going around.

11.4 Six computed witnesses

The exploratory work recorded in [5] produced six separately computed statements about punctured configurations. They are reproduced here in the register's own order, each with its exact content and its status. None of them is promoted by appearing in this paper: the register carries no authoritative status, and the items below are classified according to the labels of this repository.

1. *A topological witness.* Cut a disc carrying four right-angle sectors along the two diagonal seams and the four coordinate rays; eight triangular pieces result, each with a left radial edge, an outer arc, and a right radial edge. Regluing is a permutation σ of the eight left edges onto the eight right edges. If C is the number of cycles of σ , then the centre vertex has C orbits, the outer vertices form eight orbits, and $V = C + 8$, $E = 16$, $F = 8$, whence

$$\chi = V - E + F = C = b,$$

the number of boundary components. Consequently every gluing of this family produces either one disc or a disjoint union of C discs; a three-holed surface ($\chi = -1$, $b = 3$) is unreachable,

and equals a disc with two points removed. An exhaustive check of all 40320 permutations confirms the distribution 5040, 13068, 13132, 6769, 1960, 322, 28, 1 for $C = 1, \dots, 8$. *Status*: the identity $\chi = C$ is elementary and provable as above; the enumeration is a computationally verified example.

2. *A combinatorial witness*. If three components interact pairwise, the interaction sites are the three unordered pairs, and each site carries two directions of exchange, giving six directed transitions. In the twelve-state cycle of the next witness each directed transition occurs four times. *Status*: counting, given the definition of a hole as an interaction site. The definition is a choice, not a theorem.
3. *A frame witness*. Let the state of a three-component frame be (c, σ, φ) with component label $c \in \mathbb{Z}_3$, direction $\sigma \in \{\pm 1\}$, and phase $\varphi \in \mathbb{Z}_4$, and let one step act by

$$(c, \sigma, \varphi) \mapsto (c + \sigma, \sigma, \varphi + \sigma).$$

Since $\gcd(3, 4) = 1$, the Chinese remainder theorem identifies (c, φ) with a single counter k modulo 12; the orbit of one direction has length 12. Because one full phase period is four steps and $4 \equiv 1 \pmod{3}$, one period of the phase returns the phase to its initial value while advancing the component label by 1. The phase loop is therefore not closed in the total state: its monodromy is the generator of $\mathbb{Z}_3$. *Status*: exact arithmetic (Chinese remainder theorem), verified exhaustively on the 12-state orbit and by replay.

4. *An observer witness*. Suppose an observer sees only residuals of magnitude at least ε , and a calibration procedure stops as soon as no visible residual remains. On the frame model recorded in the register, the residuals vanish exactly at the ideal frame, and the procedure halts at $|\theta - \pi/2| = 2.83 \times 10^{-3}$ after eleven rounds for $\varepsilon = 10^{-2}$, and at 2.89×10^{-13} after fifty-six rounds for $\varepsilon = 10^{-12}$. *Status*: a property of the declared stopping rule, with the quoted numbers computationally verified. It is *not* an obstruction: nothing in the computation shows that no procedure can do better, and the exact zero of the residual at the ideal frame shows the opposite.
5. *A semantic witness*. Model the interaction sites by vocabularies: four shared terms, two terms shared by each pair, and one term private to each component. Filling the three sites independently gives $6 \cdot 5 \cdot 5 = 150$ possibilities; filling them coherently from the shared core gives $4^3 = 64$. The difference 86 is carried by the adapters at the sites. On the structural layer the pairwise intersections are non-empty while the common intersection is empty. *Status*: the arithmetic is verified given the model; the model is a modelling choice, the "Borromean" description is an analogy with no link-theoretic content, and the emptiness of the triple intersection holds for the structural layer only — the full vocabularies share the four common terms.
6. *A phase-geometry witness*. With components at phases $\{0, +1, -1\}$ modulo n , a forward motion of k steps and a reverse motion of m steps meet exactly when $k + m \equiv 0 \pmod{n}$. A meeting phase different from the three component phases exists precisely for $n \geq 4$; at $n = 4$ it is the first case in which the meeting is simultaneously balanced ($k = m$) and antipodal ($\varphi = n/2$), namely $k = m = 2$, $\varphi = 2$. *Status*: computationally verified example over the stated finite model, exhaustive for $n = 3, \dots, 8$. The value $n = 4$ is minimal, not unique: $n = 6$ and $n = 8$ also admit balanced antipodal meetings.

Remark 11.3 (Register bookkeeping). The exploration register introduces these items as five classes while tabulating six rows; the sixth row is the phase-geometry witness listed last above. This paper follows the table and counts six, recording the discrepancy rather than resolving it silently [5].

11.5 What the witnesses do not establish

The register's own summary sentence is that a hole "is not a defect to be removed but the position that carries exchange". That sentence is the motivation for this chapter, and it is not proved by any of the six witnesses. The following list of non-claims is part of the content of this section.

- No witness shows that punctures are unavoidable in AES. The witnesses are about chosen finite models: a chosen gluing family, a chosen frame, a chosen vocabulary, a chosen stopping rule. Changing the model changes the count, and the register says so for its own semantic witness.
- No witness establishes an obstruction of AEG. The monodromy of the frame witness is a monodromy of a stipulated finite flow; the calibration witness is a property of a stipulated halting rule; the gluing witness is a statement about one gluing family.
- The $\mathbb{Z}_3$ monodromy of the frame witness is not identified with any $\mathbb{Z}_4$ twist of sheet labels appearing elsewhere in the register. Those are different groups, and no computation in the register relates them; the identification asserted there is recorded as unproved.
- The word "hole" in this chapter means a puncture in the sense of Definition 11.1, or an interaction site in the sense of the combinatorial witness. It is not the "one-hole context" of Section 2, which is a placeholder in an expression, and the two must not be conflated.
- The engineering study that accompanies the register is explicitly recorded by its author as an analogy and not as a mathematical assertion; one of its three language prototypes was never compiled. It contributes motivation only.
- Nothing here changes the status of Paper I's regular-zero theorems or of Paper III's singular theory. A punctured AES is a specialization of the singular notion, and its classification is not attempted.

11.6 Open problems

The register closes with the author's own verdict that this route is not yet free, for three reviewable reasons: the family of cuts depends on a chosen permutation, the covering sheets are defined by the cut, and the graded description depends on an externally supplied frame. It registers three directions that have not been attempted, and they are the natural open problems of this chapter.

1. Replace the tangent cone of Proposition 11.2 by the formal neighbourhood of the puncture: compute the successive jets at p and test whether the description ceases to depend on the choice of tangent directions.
2. Blow up the puncture and rewrite the cut-and-glue construction as a canonical construction on the exceptional divisor, so that the seam is replaced by a construction the data itself determines.
3. Replace the cut-and-glue calculus by an operadic composition, in which the gluing of punctured surfaces is intrinsic and contains no chosen seam.

Each of these is a programme, not a result. Related and independent problems were already recorded at the end of the earlier prelude: whether a punctured model carries a ripple structure that survives at the puncture, and whether the choice of punctures can be made canonical rather than stipulated. A solution of either would turn the language of this chapter from a description into a theory.

12 Interfaces to Papers I–IV

This paper was enlarged to become the geometric foundation of the series. The division of labour between Paper 0 and Paper I is fixed by the authoritative amendment ‘governance/00b-paper-0-geometric-foundation-amendment.md’, and this section records the resulting interface without restating either paper’s theorems.

12.1 What Paper 0 establishes

The proved spine of this paper is

left and right comb grammars $\longrightarrow$ slot-(1)/slot-(2) one-hole maps,
distributive expansion $\longrightarrow$ same operator, different charges,
arithmetic motions on an AES $\longrightarrow$ point path in $\mathfrak{E}_0$,
projective operator $\longrightarrow$ pole, fixed points, ripple pencil,
affine cocycles $\longrightarrow$ relative defect $\longrightarrow$ ACS weighted area,
contact form $\longrightarrow$ non-integrable horizontal distribution $\longrightarrow$ curvature $\mu\lambda$,
 $\text{Aff}(1, K) = \text{Stab}(\infty) \subset \text{PGL}_2(K)$.

The paper also supplies four laboratories: the basic hyperbolic model $\mathfrak{E}_0$, the one-puncture model $\mathfrak{E}_1$, the pulled-back horocycle pencil, and the three-dimensional contact model. The finite geometric-series and continued-fraction examples are exact at every truncation; their infinite values are asserted only under the stated formal or analytic convergence hypotheses.

Two statements in the paper are not theorems about AES and are labelled as proposals where they occur: the descriptive name *ripple geometry* for the dual reading of Section 4, and the language of *tearing* of Section 8 together with the punctured reading of Section 11. Both rest on proved computations — the line-to-circle theorem, the weighted torsion–Stokes theorem, and the contact curvature — and neither claims a new isomorphism class of geometry.

12.2 What remains in Paper I

Paper I begins one level earlier and keeps the history layer. It retains responsibility for:

1. the intrinsic dependency-poset classification of all sequential binary trees, including mixed slot words;
2. marked planar sequential trees and bounded/free marked spinal histories, with operand-slot chirality, mirror, temporal reversal, and path inverse kept distinct;
3. projective evaluation of chronological marked histories, including ordinary-domain data and the affine sector;
4. the $q = 4$ Hecke sublanguage and its history-level relations;

5. the regular-zero theorems, the complete splitting theorem, the foundational definition of a singular arithmetic expression space, the verified isolated-zero model, and the smooth parameter-family zero-set lemma;
6. the interfaces to Papers II, III, and IV.

Paper I states the definition of a regular AES and its canonical frame as a concise imported interface, because its zero theorems are stated in those terms; everything else that moved here is referenced rather than repeated. The earlier version of that interface, which ran in the opposite direction, is superseded.

12.3 The dependency architecture

The series dependency is

$$\boxed{\text{Paper 0} \longrightarrow \text{Paper I} \longrightarrow \{\text{Papers II, III, IV}\}.$$

The first arrow exports the following interfaces:

Paper 0 object	use in later papers
comb and slot conventions	operand-slot chirality on sequential spines
distributive expansion	comparison of expansions with equal operator
regular AES and canonical frame	imported definition for zero theory
arithmetic motions and $\mathfrak{E}_0$	model for analysis and for the flow
one-puncture model $\mathfrak{E}_1$	first punctured instance
point path	bounded marked-history evaluation in the affine sector
ripple pencil and pole	motivation for bilateral projective continuation
affine cocycles	discrete order defect
ACS and relative torsion	global torsion and its measures
contact form and horizontal curvature	horizontal calculus and function theory
tearing and punctured spaces	vocabulary for singular and resource questions

Papers II and III import the regular-AES definition, the canonical frame, the basic hyperbolic model, the contact model, and the elementary bilateral operators from this paper; they do not redefine them. Paper IV imports the history layer from Paper I and the affine torsion from this paper.

12.4 Open extensions

Several genuine problems remain, and they are recorded here rather than advertised as results.

1. Construct a projective analogue of the affine arithmetic expression space in which the pole and the ripple pencil are intrinsic rather than chosen through one upper-half-plane chart, and decide which path or ripple data descend through relations among history words.

2. Decide whether the contact structure of Section 9 admits a canonical compatible structure beyond the horizontal distribution, and if so which. Neither a complex structure nor a horizontal metric is determined by the contact form alone.
3. Determine whether the choice of punctures in Section 11 can be made canonical, and whether a punctured model carries a ripple structure that survives at the puncture; the three registered programmes of Subsection 11.6 are the concrete routes.
4. Extend the one-hole analysis to multi-wire expression networks while retaining ordinary-domain, pole, and fixed-point information, and develop a controlled theory of infinite arithmetic expressions that separates formal, analytic, projective, and computational convergence.
5. Determine whether the non-linear algebraic structures proposed in Section 10 can be defined intrinsically for AES, so that the two proved obstructions of that section are replaced by a positive description rather than a refusal.

None of these problems is solved merely by the matrix representation or by the existence of the contact form. The stable conclusion of the present paper is more elementary: the four arithmetic operations act as motions of one space, their order defect is measured in the commutativization of that space, and the contact structure of the same defect is what prevents the two motions from being calibrated independently.

A Elementary Calculations

This appendix records calculations used in the main text. They are included to keep the elementary claims reproducible and to fix chronology and sign conventions.

A.1 Chronological products and one-hole recursion

Let

$$F_k(z) = \frac{A^{(k)}z + B^{(k)}}{C^{(k)}z + D^{(k)}}$$

have matrix representative

$$M_k = \begin{pmatrix} A^{(k)} & B^{(k)} \\ C^{(k)} & D^{(k)} \end{pmatrix}.$$

If F_1 acts first, the two-step action is

$$F_2(F_1(z)),$$

and direct substitution gives the matrix M_2M_1 . Induction therefore gives

$$F_k \circ \cdots \circ F_1 \longleftrightarrow M_k \cdots M_1.$$

This is the convention used in (44).

For affine matrices

$$M_k = \begin{pmatrix} \alpha_k & \beta_k \\ 0 & 1 \end{pmatrix}, \quad P_{k-1} = \begin{pmatrix} A_{k-1} & B_{k-1} \\ 0 & 1 \end{pmatrix},$$

one has

$$M_k P_{k-1} = \begin{pmatrix} \alpha_k A_{k-1} & \alpha_k B_{k-1} + \beta_k \\ 0 & 1 \end{pmatrix},$$

which is the recursion (20).

A.2 The ripple circle calculation

For

$$F(z) = \frac{Az + B}{Cz + D}, \quad \Delta = AD - BC > 0,$$

one computes

$$\begin{aligned} F(z) - \overline{F(z)} &= \frac{(Az + B)(C\bar{z} + D) - (A\bar{z} + B)(Cz + D)}{|Cz + D|^2} \\ &= \frac{(AD - BC)(z - \bar{z})}{|Cz + D|^2}. \end{aligned}$$

Dividing by $2i$ gives

$$\operatorname{Im} F(z) = \frac{\Delta \operatorname{Im} z}{|Cz + D|^2}.$$

Writing $z = x + iy$ and imposing $\operatorname{Im} F(z) = t$ gives

$$\Delta y = t((Cx + D)^2 + C^2 y^2).$$

When $C \neq 0$, division by tC^2 yields

$$\left(x + \frac{D}{C}\right)^2 + y^2 - \frac{\Delta}{tC^2}y = 0,$$

and completing the square yields (32).

For $J(z) = -1/z$, one has $A = D = 0$, $B = -1$, $C = 1$, and $\Delta = 1$. Hence

$$x^2 + \left(y - \frac{1}{2t}\right)^2 = \left(\frac{1}{2t}\right)^2.$$

A.3 Powers of an affine map

Let $T(z) = az + b$ with $a \neq 0$. Matrix multiplication gives

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} a^n & b(1 + a + \cdots + a^{n-1}) \\ 0 & 1 \end{pmatrix}.$$

If $a \neq 1$, this is

$$\begin{pmatrix} a^n & b(1 - a^n)/(1 - a) \\ 0 & 1 \end{pmatrix}.$$

Writing $q = b/(1 - a)$, the induced map becomes

$$T^n(z) = a^n z + q(1 - a^n) = q + a^n(z - q).$$

If $a = 1$, the matrix power is

$$\begin{pmatrix} 1 & nb \\ 0 & 1 \end{pmatrix},$$

so $T^n(z) = z + nb$.

For the geometric-series update $T_r(s) = rs + 1$, apply the same formula with $a = r$, $b = 1$, and $s = 0$:

$$T_r^n(0) = \frac{1 - r^n}{1 - r}$$

when $r \neq 1$.

A.4 Continued-fraction matrices

Let

$$F_k(z) = a_k + \frac{b_k}{z}, \quad M_k = \begin{pmatrix} a_k & b_k \\ 1 & 0 \end{pmatrix}, \quad b_k \neq 0.$$

Write

$$P_k = M_k \cdots M_1 = \begin{pmatrix} p_k & r_k \\ q_k & s_k \end{pmatrix}.$$

Then

$$P_k = \begin{pmatrix} a_k & b_k \\ 1 & 0 \end{pmatrix} \begin{pmatrix} p_{k-1} & r_{k-1} \\ q_{k-1} & s_{k-1} \end{pmatrix} = \begin{pmatrix} a_k p_{k-1} + b_k q_{k-1} & a_k r_{k-1} + b_k s_{k-1} \\ p_{k-1} & r_{k-1} \end{pmatrix}.$$

Thus

$$q_k = p_{k-1}, \quad p_k = a_k p_{k-1} + b_k p_{k-2},$$

with the analogous recurrence for the second column. The k th truncation at an initial value z_0 is

$$F_k \circ \cdots \circ F_1(z_0) = \frac{p_k z_0 + r_k}{q_k z_0 + s_k}.$$

An ordinary truncation is admissible precisely when each intermediate denominator is nonzero; the projective formula also records the stages at which the orbit reaches infinity.

In the stationary golden-ratio case $a_k = b_k = 1$, the first-column recurrence is

$$p_k = p_{k-1} + p_{k-2},$$

which is the Fibonacci recurrence. This gives the ratios displayed in Example 5.4.

A.5 The constructive projective factorization

For completeness, assume $C \neq 0$ and put

$$q = \frac{D}{C}, \quad s = \frac{AD - BC}{C^2}, \quad t = \frac{A}{C}.$$

Then

$$\begin{aligned} T_t \circ D_s \circ J \circ T_q(z) &= t - \frac{s}{z + q} \\ &= \frac{t(z + q) - s}{z + q} \\ &= \frac{(A/C)z + AD/C^2 - (AD - BC)/C^2}{z + D/C} \\ &= \frac{(A/C)z + B/C}{z + D/C} \\ &= \frac{Az + B}{Cz + D}. \end{aligned}$$

This verifies (46) with the signs used in the main text.

B Hyperbolic-Model Calculations

This appendix records the coordinate checks used for the basic hyperbolic arithmetic expression space. Throughout, $\mu, \lambda > 0$,

$$G = \{(x, y) : x \in \mathbb{R}, y > 0\}, \quad (x, y)(x', y') = (x + yx', yy'),$$

and

$$g_{\mu, \lambda} = \frac{1}{y^2} \left(\frac{dx^2}{\mu^2} + \frac{dy^2}{\lambda^2} \right), \quad a = -\frac{x}{y}.$$

The elementary grid-map and Baumslag–Solitar calculations formerly included here are now owned by Paper 0.

B.1 Group coordinates and the invariant frame

The identity of G is $(0, 1)$, and

$$(x, y)^{-1} = \left(-\frac{x}{y}, \frac{1}{y} \right).$$

For $g_0 = (x_0, y_0)$, left translation and its differential are

$$L_{g_0}(x, y) = (x_0 + y_0x, y_0y), \quad dL_{g_0}(\partial_x) = y_0\partial_x, \quad dL_{g_0}(\partial_y) = y_0\partial_y.$$

It follows that $y\partial_x$ and $y\partial_y$ are left-invariant. The normalized arithmetic frame

$$X_u = -\mu y \partial_x, \quad X_v = -\lambda y \partial_y$$

is therefore left-invariant, and its dual coframe is

$$-\frac{dx}{\mu y}, \quad -\frac{dy}{\lambda y}.$$

The sum of their squares is $g_{\mu, \lambda}$. The bracket is

$$\begin{aligned} [X_u, X_v] &= [-\mu y \partial_x, -\lambda y \partial_y] \\ &= -\mu \lambda y \partial_x = \lambda X_u, \end{aligned}$$

and hence $[X_u, X_v]a = \mu\lambda$, as required by the general arithmetic-frame identity.

The same invariance can be checked directly:

$$\begin{aligned} L_{(x_0, y_0)}^* g_{\mu, \lambda} &= \frac{1}{(y_0 y)^2} \left(\frac{(y_0 dx)^2}{\mu^2} + \frac{(y_0 dy)^2}{\lambda^2} \right) \\ &= g_{\mu, \lambda}. \end{aligned}$$

For a horocyclic chart, set

$$u = x, \quad y = e^{\lambda v}.$$

Then

$$(u, v)(u', v') = (u + e^{\lambda v} u', v + v'),$$

and

$$X_u = -\mu e^{\lambda v} \partial_u, \quad X_v = -\partial_v,$$

while

$$g_{\mu, \lambda} = e^{-2\lambda v} \frac{du^2}{\mu^2} + dv^2, \quad a = -ue^{-\lambda v}.$$

The letters u, v in this chart are genuine coordinates, but the arithmetic frame is not the coordinate frame.

B.2 Inverse metric, gradient, and the eikonal identity

In the (x, y) -chart,

$$(g_{ij}) = \begin{pmatrix} \frac{1}{\mu^2 y^2} & 0 \\ 0 & \frac{1}{\lambda^2 y^2} \end{pmatrix}, \quad (g^{ij}) = \begin{pmatrix} \mu^2 y^2 & 0 \\ 0 & \lambda^2 y^2 \end{pmatrix},$$

and

$$\det(g_{ij}) = \frac{1}{\mu^2 \lambda^2 y^4}, \quad d\text{vol}_g = \frac{dx \wedge dy}{\mu \lambda y^2}.$$

Since

$$a_x = -\frac{1}{y}, \quad a_y = \frac{x}{y^2} = -\frac{a}{y},$$

the gradient is

$$\begin{aligned} \nabla a &= g^{xx} a_x \partial_x + g^{yy} a_y \partial_y \\ &= -\mu^2 y \partial_x + \lambda^2 x \partial_y. \end{aligned}$$

Its squared norm is

$$\begin{aligned} |\nabla a|_g^2 &= g^{xx} a_x^2 + g^{yy} a_y^2 \\ &= \mu^2 + \lambda^2 \frac{x^2}{y^2} \\ &= \mu^2 + \lambda^2 a^2. \end{aligned}$$

Alternatively,

$$X_u a = (-\mu y) \left(-\frac{1}{y} \right) = \mu, \quad X_v a = (-\lambda y) \left(\frac{x}{y^2} \right) = \lambda a,$$

so the framed and eikonal verifications agree.

B.3 Curvature and completeness

Two calculations give the curvature normalization. First set

$$X = \frac{\lambda}{\mu} x.$$

This is a global diffeomorphism of upper half-planes, and

$$g_{\mu, \lambda} = \frac{1}{\lambda^2} \frac{dX^2 + dy^2}{y^2}.$$

The standard metric $(dX^2 + dy^2)/y^2$ is complete with curvature -1 . Constant rescaling by λ^{-2} gives a complete metric of curvature $-\lambda^2$.

For a direct Cartan calculation, use the positively oriented orthonormal coframe

$$\omega^1 = \frac{dx}{\mu y}, \quad \omega^2 = \frac{dy}{\lambda y}.$$

Then

$$d\omega^1 = \lambda \omega^1 \wedge \omega^2, \quad d\omega^2 = 0.$$

The Levi–Civita connection form $\omega^1_2 = -\lambda\omega^1$ satisfies

$$d\omega^1 = -\omega^1_2 \wedge \omega^2, \quad d\omega^2 = \omega^1_2 \wedge \omega^1.$$

Its curvature form is

$$d\omega^1_2 = -\lambda d\omega^1 = -\lambda^2 \omega^1 \wedge \omega^2.$$

Thus $K = -\lambda^2$, consistently with the coordinate rescaling.

B.4 The Laplace–Beltrami operator

With the divergence-of-gradient convention,

$$\Delta_g f = \frac{1}{\sqrt{\det g}} \partial_i \left(\sqrt{\det g} g^{ij} \partial_j f \right).$$

Here

$$\sqrt{\det g} = \frac{1}{\mu\lambda y^2}, \quad \sqrt{\det g} g^{xx} = \frac{\mu}{\lambda}, \quad \sqrt{\det g} g^{yy} = \frac{\lambda}{\mu}.$$

The last two coefficients are constant, so

$$\Delta_{g_{\mu,\lambda}} f = \mu^2 y^2 f_{xx} + \lambda^2 y^2 f_{yy}. \quad (87)$$

For $a = -x/y$,

$$a_{xx} = 0, \quad a_{yy} = -\frac{2x}{y^3} = \frac{2a}{y^2}.$$

Equation (87) therefore gives

$$\Delta_{g_{\mu,\lambda}} a = 2\lambda^2 a.$$

C Affine Cocycle Calculations

This appendix supplements Proposition 7.1 with expanded calculations. All discrete formulas hold over an arbitrary field K , and every multiplier is assumed nonzero.

C.1 Prefix recursion and closed form

For the prefix history $(f_1, \dots, f_k)$, write

$$F_k = f_k \circ \dots \circ f_1, \quad F_k(x) = \Phi_k x + \xi_k,$$

with $F_0(x) = x$. If $f_{k+1}(x) = s_{k+1}x + t_{k+1}$, then

$$\Phi_{k+1} = s_{k+1}\Phi_k, \quad \xi_{k+1} = s_{k+1}\xi_k + t_{k+1}. \quad (88)$$

Iterating the first recursion gives

$$\Phi_k = s_k \cdots s_1.$$

Iterating the second one gives

$$\begin{aligned} \xi_1 &= t_1, \\ \xi_2 &= s_2 t_1 + t_2, \\ \xi_3 &= s_3 s_2 t_1 + s_3 t_2 + t_3, \\ \xi_4 &= s_4 s_3 s_2 t_1 + s_4 s_3 t_2 + s_4 t_3 + t_4. \end{aligned}$$

The pattern is

$$\xi_k = \sum_{i=1}^k t_i \prod_{j=i+1}^k s_j.$$

Substitution in (88) proves this formula inductively, including the final term through the empty-product convention.

The source-normalized prefix coordinate $\hat{\xi}_k = \Phi_k^{-1} \xi_k$ obeys a different recursion. From (88),

$$\hat{\xi}_{k+1} = \frac{s_{k+1} \xi_k + t_{k+1}}{s_{k+1} \Phi_k} = \hat{\xi}_k + \frac{t_{k+1}}{\Phi_{k+1}}. \quad (89)$$

Starting from $\hat{\xi}_0 = 0$ therefore gives

$$\hat{\xi}_k = \sum_{i=1}^k \frac{t_i}{\Phi_i} = \sum_{i=1}^k \frac{t_i}{\prod_{j=1}^i s_j}.$$

This second derivation makes the past normalization explicit at every prefix.

C.2 Suffix transport and the future-weight kernel

For $0 \leq i \leq n$, define the scale of the strict suffix after step i by

$$S_{i \rightarrow n} = \prod_{j=i+1}^n s_j, \quad S_{n \rightarrow n} = 1.$$

Then the target-frame translation is simply

$$\xi_n = \sum_{i=1}^n S_{i \rightarrow n} t_i. \quad (90)$$

To verify the interpretation, replace only t_i by $t_i + \epsilon$. All states before step i are unchanged. At step i the output gains ϵ , and every later affine map multiplies this change by its scale. Thus the final endpoint changes by

$$\epsilon s_{i+1} \cdots s_n = \epsilon S_{i \rightarrow n}.$$

The coefficient in (90) is therefore exactly the transport of an additive event to the target frame.

C.3 Alternating addition–scale histories

Consider a positive real alternating history with pairs

$$x \xrightarrow{e^{m_i}} e^{m_i} x \xrightarrow{+\mu_i} e^{m_i} x + \mu_i, \quad i = 1, \dots, n,$$

applied from $i = 1$ to $i = n$. At the level of affine steps this has $s_i = e^{m_i}$ and $t_i = \mu_i$. Define prefix and suffix logarithmic scales by

$$M_i = \sum_{j=1}^i m_j, \quad M_{i+1:n} = \sum_{j=i+1}^n m_j.$$

Proposition 7.1 becomes

$$\Phi_n = e^{M_n}, \quad (91)$$

$$\xi_n = \sum_{i=1}^n \mu_i e^{M_{i+1:n}}, \quad (92)$$

$$\hat{\xi}_n = \sum_{i=1}^n \mu_i e^{-M_i}. \quad (93)$$

For $n = 3$, direct expansion checks the target formula:

$$\begin{aligned} x &\mapsto e^{m_1}x + \mu_1 \\ &\mapsto e^{m_2+m_1}x + e^{m_2}\mu_1 + \mu_2 \\ &\mapsto e^{m_3+m_2+m_1}x + e^{m_3+m_2}\mu_1 + e^{m_3}\mu_2 + \mu_3. \end{aligned}$$

Dividing the translation line by $e^{m_1+m_2+m_3}$ gives

$$\hat{\xi}_3 = \mu_1 e^{-m_1} + \mu_2 e^{-(m_1+m_2)} + \mu_3 e^{-(m_1+m_2+m_3)},$$

in agreement with (93).

The convention in this example is “scale, then add” within each pair. If a source instead groups the elementary operations as “add, then scale”, the translation attached to that pair is $e^{m_i}\mu_i$, not μ_i . Converting the pair to the common affine form $s_i x + t_i$ before using the cocycle formulas prevents this convention change from being hidden.

C.4 Concatenation in target and normalized coordinates

Let

$$\rho(\gamma) = g(\Phi_\gamma, \xi_\gamma), \quad \rho(\delta) = g(\Phi_\delta, \xi_\delta),$$

where $g(\Phi, \xi)(x) = \Phi x + \xi$. Since $\gamma\delta$ performs γ first,

$$\begin{aligned} \rho(\gamma\delta)(x) &= \rho(\delta)(\rho(\gamma)(x)) \\ &= \Phi_\delta(\Phi_\gamma x + \xi_\gamma) + \xi_\delta \\ &= \Phi_\delta \Phi_\gamma x + \Phi_\delta \xi_\gamma + \xi_\delta. \end{aligned}$$

This yields the target-coordinate pair

$$(\Phi_{\gamma\delta}, \xi_{\gamma\delta}) = (\Phi_\delta \Phi_\gamma, \Phi_\delta \xi_\gamma + \xi_\delta).$$

For normalized coordinates,

$$\begin{aligned} \hat{\xi}_{\gamma\delta} &= \frac{\Phi_\delta \xi_\gamma + \xi_\delta}{\Phi_\delta \Phi_\gamma} \\ &= \hat{\xi}_\gamma + \Phi_\gamma^{-1} \hat{\xi}_\delta. \end{aligned}$$

The first formula transports the earlier translation forward by the later scale. The second transports the later normalized translation back through the earlier scale.

For three histories, repeated use gives

$$\begin{aligned} \xi_{\gamma\delta\epsilon} &= \Phi_\epsilon \Phi_\delta \xi_\gamma + \Phi_\epsilon \xi_\delta + \xi_\epsilon, \\ \hat{\xi}_{\gamma\delta\epsilon} &= \hat{\xi}_\gamma + \Phi_\gamma^{-1} \hat{\xi}_\delta + (\Phi_\delta \Phi_\gamma)^{-1} \hat{\xi}_\epsilon. \end{aligned}$$

These are the block-history versions of future weighting and past normalization.

C.5 Inverse histories and relative comparison

The affine inverse is

$$g(\Phi, \xi)^{-1} = g(\Phi^{-1}, -\Phi^{-1}\xi). \quad (94)$$

Indeed,

$$g(\Phi^{-1}, -\Phi^{-1}\xi)(\Phi x + \xi) = x + \Phi^{-1}\xi - \Phi^{-1}\xi = x.$$

Using (94), the relative affine defect is

$$\rho(\delta)^{-1}\rho(\gamma) = g\left(\frac{\Phi_\gamma}{\Phi_\delta}, \frac{\xi_\gamma - \xi_\delta}{\Phi_\delta}\right). \quad (95)$$

In matrices, the same computation is

$$\begin{pmatrix} \Phi_\delta^{-1} & -\Phi_\delta^{-1}\xi_\delta \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \Phi_\gamma & \xi_\gamma \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \Phi_\delta^{-1}\Phi_\gamma & \Phi_\delta^{-1}(\xi_\gamma - \xi_\delta) \\ 0 & 1 \end{pmatrix}.$$

If $\Phi_\gamma = \Phi_\delta = \Phi$, the relative element is a pure translation by $(\xi_\gamma - \xi_\delta)/\Phi$. The unnormalized endpoint difference is $\xi_\gamma - \xi_\delta$. These two scalar quantities differ by the common scale and should not be identified.

C.6 The elementary order square

Let

$$A_\mu = \begin{pmatrix} 1 & \mu \\ 0 & 1 \end{pmatrix}, \quad S_q = \begin{pmatrix} q & 0 \\ 0 & 1 \end{pmatrix}.$$

Under the chronological convention, first adding and then scaling has matrix

$$S_q A_\mu = \begin{pmatrix} q & q\mu \\ 0 & 1 \end{pmatrix},$$

whereas first scaling and then adding has matrix

$$A_\mu S_q = \begin{pmatrix} q & \mu \\ 0 & 1 \end{pmatrix}.$$

Their endpoint difference is

$$(qx + q\mu) - (qx + \mu) = \mu(q - 1).$$

Taking the second history as reference gives

$$(A_\mu S_q)^{-1}(S_q A_\mu) = \begin{pmatrix} 1 & \mu(1 - q^{-1}) \\ 0 & 1 \end{pmatrix}.$$

Thus the relative group translation is $\mu(1 - q^{-1})$, while the open endpoint defect is $\mu(q - 1)$. The latter is the former multiplied by the common target scale q .

Closing the comparison into a group commutator produces yet another ordered quantity. For example,

$$\begin{aligned} S_q A_\mu S_q^{-1} A_\mu^{-1} &= \begin{pmatrix} 1 & (q - 1)\mu \\ 0 & 1 \end{pmatrix}, \\ A_\mu S_q A_\mu^{-1} S_q^{-1} &= \begin{pmatrix} 1 & (1 - q)\mu \\ 0 & 1 \end{pmatrix}. \end{aligned}$$

The signs differ because the loop orientations differ. These closed-loop calculations are recorded only to audit conventions; the torsion theory and its orientation choices are introduced in Section 8.

C.7 Exact perturbation propagation at every prefix

Let x and $\tilde{x}$ be two initial values, and propagate both through the same history. Set

$$x_k = F_k(x), \quad \tilde{x}_k = F_k(\tilde{x}).$$

From the prefix recursion,

$$\begin{aligned} \tilde{x}_{k+1} - x_{k+1} &= (s_{k+1}\tilde{x}_k + t_{k+1}) - (s_{k+1}x_k + t_{k+1}) \\ &= s_{k+1}(\tilde{x}_k - x_k). \end{aligned}$$

Induction gives

$$\tilde{x}_k - x_k = \left(\prod_{i=1}^k s_i \right) (\tilde{x} - x) = \Phi_k(\tilde{x} - x). \quad (96)$$

The translations cancel at each step; only the scale cocycle propagates the initial difference. Over $\mathbb{R}$, if $s_i = e^{\lambda_i}$, then

$$\frac{\tilde{x}_k - x_k}{\tilde{x} - x} = e^{\lambda_1 + \dots + \lambda_k}.$$

No limiting process is involved in this identity.

C.8 Differential conversion check

Finally, on the positive real affine group let $\Phi = e^M$ and $\hat{\xi} = e^{-M}\xi$. The product rule gives

$$\begin{aligned} d\hat{\xi} &= d(e^{-M}\xi) \\ &= -e^{-M}\xi dM + e^{-M}d\xi \\ &= e^{-M}(d\xi - \xi dM). \end{aligned}$$

Therefore

$$d\xi - \xi dM = e^M d\hat{\xi}.$$

This verifies that the translation component of $dg g^{-1}$ is the target scale times the differential of the source-normalized coordinate. It does not identify $dg g^{-1}$ with $g^{-1}dg$, whose translation component is $e^{-M}d\xi$.

C.9 Left and right group data

The affine group is represented by

$$g(\Phi, \xi) = \begin{pmatrix} \Phi & \xi \\ 0 & 1 \end{pmatrix}, \quad \Phi \in K^\times.$$

With the column action, matrix multiplication gives

$$g(\Phi_2, \xi_2)g(\Phi_1, \xi_1) = g(\Phi_2\Phi_1, \Phi_2\xi_1 + \xi_2). \quad (97)$$

Thus the earlier translation is rescaled by the later affine map. Equation (97) is the two-factor origin of the target-frame cocycle.

Over the positive real component, write $\Phi = e^M$. The two Maurer–Cartan forms used in Section 7 have the following readings.

Form	Translation component	Frame
$g^{-1}dg$	$e^{-M}d\xi$	body/source
$dg g^{-1}$	$d\xi - \xi dM$	spatial/target

The words “left” and “right” here refer to invariant differential forms on the affine group. They do not refer to first and second operand slots. The operand-slot mirror is syntactic; the Maurer–Cartan distinction is group-theoretic.

C.10 Target and source-normalized translation coordinates

For $x \mapsto \Phi x + \xi$, the symbol ξ always denotes target-frame translation. The source-normalized coordinate is

$$\widehat{\xi} = \Phi^{-1}\xi.$$

Their discrete and differential conversions are

$$\xi = \Phi \widehat{\xi}, \quad \widehat{\xi} = \Phi^{-1}\xi,$$

and, when $\Phi = e^M$,

$$d\xi = \xi dM + e^M d\widehat{\xi}.$$

No formula in the main text changes from ξ to $\widehat{\xi}$ without this normalization being displayed.

D ACS and Contact Calculations

This appendix records sign checks and coordinate calculations used in Sections 8 and 9. The chronological convention is unchanged: entries in a history are performed from left to right. The ACS is oriented by $dA \wedge dM$, and the contact state space is oriented by $du \wedge da \wedge dv$.

D.1 The oriented ACS rectangle

Let

$$\gamma = (A_p, M_q), \quad \delta = (M_q, A_p).$$

Their common terminal multiplicative charge is $M_* = q$. The chain $C_\gamma - C_\delta$ traverses the bottom edge from $(0,0)$ to $(p,0)$, the right edge to (p,q) , the top edge back to $(0,q)$, and the left edge back to the origin. For $p, q > 0$ this is the positive boundary convention for $dA \wedge dM$. Since vertical edges do not contribute to $\eta_* = e^{q-M}dA$,

$$\begin{aligned} \int_{C_\gamma - C_\delta} \eta_* &= \int_0^p e^q dA + \int_p^0 1 dA \\ &= p(e^q - 1). \end{aligned}$$

On the other hand,

$$\int_{[0,p] \times [0,q]} e^{q-M} dA \wedge dM = p \int_0^q e^{q-M} dM = p(e^q - 1).$$

This verifies both the sign and the target-frame weight.

The same formula holds for signed p and q . More precisely, regard the rectangle as the oriented singular chain parametrized by

$$R_{p,q}(s,t) = (sp, tq), \quad 0 \leq s, t \leq 1.$$

Its pulled-back orientation form is $pq \, ds \wedge dt$, so orientation reverses automatically when exactly one side is negative. Direct integration still gives

$$\int_{R_{p,q}} e^{q-M} dA \wedge dM = p(e^q - 1).$$

Thus no separate “unsigned area” convention is needed.

D.2 A closed charge path with nonzero evaluation defect

Consider the chronological loop

$$\ell = (A_p, M_q, A_{-p}, M_{-q}).$$

Its total ACS charge is $(0, 0)$, the same as the empty history $\emptyset$, but

$$\begin{aligned} \nu_x(\ell) &= e^{-q}(e^q(x+p) - p) \\ &= x + p(1 - e^{-q}). \end{aligned}$$

Hence

$$\tau(\ell, \emptyset) = p(1 - e^{-q}).$$

Here $M_* = 0$, so $\eta_* = e^{-M} dA$. The ACS path of ℓ is the oriented rectangle boundary itself, and

$$\int_{C_\ell} e^{-M} dA = p - pe^{-q} = p(1 - e^{-q}).$$

This example simultaneously checks the closed-loop sign and shows why zero total charge is weaker than operator neutrality.

For a self-intersecting charge path, the same computation is performed on a singular two-chain with integer multiplicities. Decomposing a filling into oriented rectangles adds their signed weighted integrals; overlap is counted with the sum of multiplicities. This is the precise replacement for speaking of “the region enclosed” by a self-intersecting loop.

D.3 Finite contact compositions

Write $p = \mu h$ and $q = \lambda k$. On the assignment coordinate, the two horizontal flows act by

$$A_h(a) = a + p, \quad M_k(a) = e^q a.$$

The open comparison is

$$\begin{aligned} (M_k \circ A_h)(a) &= e^q(a + p), \\ (A_h \circ M_k)(a) &= e^q a + p, \end{aligned}$$

and therefore

$$(M_k \circ A_h)(a) - (A_h \circ M_k)(a) = p(e^q - 1).$$

For the closed chronological word A_h, M_k, A_{-h}, M_{-k} , the induced composition is

$$M_{-k} \circ A_{-h} \circ M_k \circ A_h,$$

and

$$\begin{aligned} (M_{-k} \circ A_{-h} \circ M_k \circ A_h)(a) &= e^{-q}(e^q(a+p) - p) \\ &= a + p(1 - e^{-q}). \end{aligned}$$

Consequently

$$T_{\text{closed}} = p(1 - e^{-q}) = e^{-q}p(e^q - 1) = e^{-q}T_{\text{open}}.$$

The equality after multiplication by e^q is a change to the common target frame, not an equality of the two raw finite defects.

D.4 Darboux coordinates and the Reeb field

Assume $\mu\lambda \neq 0$ and introduce natural coordinates

$$\tilde{u} = \mu u, \quad \tilde{v} = \lambda v.$$

Then

$$\alpha = da - d\tilde{u} - a d\tilde{v}.$$

Set

$$x = \tilde{v}, \quad y = a, \quad z = a - \tilde{u}.$$

This is a global affine change of coordinates on $\mathbb{R}^3$, and

$$dz - y dx = da - d\tilde{u} - a d\tilde{v} = \alpha.$$

Thus the contact form is in the standard Darboux form. The arithmetic structure is not a new local contact-isomorphism class; it is the distinguished frame (D_u, D_v) and its interpretation by addition and scaling.

For completeness, the Reeb field R is determined by

$$\alpha(R) = 1, \quad \iota_R d\alpha = 0.$$

Since $d\alpha = -\lambda da \wedge dv$, its kernel is spanned by ∂_u . Hence

$$R = -\frac{1}{\mu}\partial_u.$$

In the Darboux coordinates above this is $R = \partial_z$, as expected.

D.5 Expanded scalar-curvature calculation

Let $F = F(u, v, a)$. Expanding the two ordered derivatives gives

$$\begin{aligned} D_u D_v F &= F_{uv} + \lambda a F_{ua} + \mu F_{av} + \mu \lambda F_a + \mu \lambda a F_{aa}, \\ D_v D_u F &= F_{vu} + \mu F_{va} + \lambda a F_{au} + \mu \lambda a F_{aa}. \end{aligned}$$

Equality of the mixed coordinate derivatives leaves

$$D_u D_v F - D_v D_u F = \mu \lambda F_a.$$

Multiplying by $du \wedge dv$ yields

$$\mathcal{R}_H(F) = \mu \lambda (\partial_a F) du \wedge dv.$$

This is an antisymmetrized calculation on scalar fields. It neither defines δ_H on one-forms nor claims that a graded differential has been constructed.

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